Geometric Prediction of Ω_Λ and r_s from $\mathbb{RP}^4$ Topology: BAO Validation with Zero Parameters Fitted to Data

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Abstract

The cosmological constant problem — why the observed dark energy density is $\sim 10^{120}$ times smaller than the naive quantum field theory prediction — has no accepted solution. We present a topological approach: if the pre-collapse state of the universe is invariant under the antipodal map $x \sim -x$, then by the Euler characteristic theorem the unique compact positive-curvature 4-manifold consistent with this symmetry is $\mathbb{RP}^4 = S^4/\mathbb{Z}_2$. The topology is a **theorem**, not an axiom.

The $\mathbb{Z}_2$ identification projects out all odd-l spherical harmonics, shifting the Fibonacci self-termination condition from $l = 1$ (BAO scale 4.4 times too large on S^4) to $l = 4$, where $d(S^4, 4) = 55 = F_{10}$. This single constraint, with no free parameters, fixes the full IHC structure: $N = 33$ nested toroidal shells, and $r_s = 153.2$ Mpc. The cosmological constant is addressed via a UV–IR Casimir seesaw $\rho_\Lambda = \sqrt{\rho_{UV}|\rho_{IR}|}$, where ρ_{IR} is fixed by $Z^{\rm reg}(-1) = -631/30$ (exact rational), giving $\Omega_\Lambda = 0.6882$. A second independent route via $\mathbb{Z}_3$ coherent interference gives $\Omega_\Lambda = 0.6889$; the 0.10% agreement between two structurally independent derivations is a non-trivial internal consistency check.

Against seven BAO surveys (33 measurements, $z = 0.106$–2.33), IHC achieves $\chi^2/n = 0.916$ versus ΛCDM's 1.196 with zero fitted parameters. Bayesian evidence via nested sampling gives $\ln B = +4.76$ (moderate evidence, Jeffreys scale) and a joint four-parameter MCMC places the IHC prediction at Mahalanobis distance 0.70σ from the posterior mean (posterior predictive p-value $= 0.61$).

Keywords: real projective space; $\mathbb{RP}^4$ topology; inversion symmetry; Euler characteristic; spherical harmonics; Fibonacci sequence; cosmological constant problem; baryon acoustic oscillations; Casimir effect; UV–IR seesaw; Bayesian model comparison; DESI DR2

1 Introduction

The topological result. We prove that $\mathbb{RP}^4 = S^4/\mathbb{Z}_2$ is not merely a modelling choice for cosmology but a *mathematical necessity* given one physical symmetry requirement. If

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the pre-collapse state of the universe is invariant under the antipodal map $x \mapsto -x$ — the inversion principle, the statement that no point was preferred over its antipode — then by the Euler characteristic theorem, the only compact positive-curvature 4-manifold consistent with this symmetry is $\mathbb{RP}^4$ (Section 2.2). The $\mathbb{Z}_2$ identification simultaneously projects out all odd-l spherical harmonics from the spectrum, a constraint with a decisive observational consequence: the Fibonacci self-termination condition on the S^4 harmonic spectrum fires at $l = 4$ rather than $l = 1$, determining the entire structure of the Inverted Hypersphere Cosmology (IHC) framework with no free parameters.

The cosmological consequence. The cosmological constant problem remains one of the most profound challenges in theoretical physics. Quantum field theory predicts a vacuum energy density $\rho_{\text{vac}}^{\text{QFT}}$ that exceeds observations by ~ 120 orders of magnitude [1, 2]. While the ΛCDM model successfully describes CMB anisotropies, large-scale structure, and cosmic expansion [3–5], it treats Ω_Λ as a free parameter fitted to data. No theoretical principle determines why $\Omega_\Lambda \approx 0.69$.

The $\mathbb{RP}^4$ topology derived above addresses this: the antipodal identification forces dark energy to the geometric mean of the UV and IR vacuum scales,

$$\rho_\Lambda = \sqrt{\rho_{UV}\,|\rho_{IR}|}$$

where ρ_{IR} is fixed by the exact rational $\mathbb{RP}^3$ Casimir spectrum ($Z^{\text{reg}}(-1) = -631/30$, no free parameters), giving $\Omega_\Lambda = 0.6882$. A second independent route via the $\mathbb{Z}_3$ counter-rotation structure gives $\Omega_\Lambda = 0.6889$; the 0.10% agreement between two structurally independent derivations is a non-trivial internal consistency check of the $\mathbb{RP}^4$ geometry.

The $l = 4$ spectral termination also determines $N = 33$ nested toroidal shells with golden-ratio spacing and $r_s^{\text{IHC}} = 153.2$ Mpc from the $k = 7$ toroidal shell. Neither Ω_Λ nor r_s is fitted to BAO or CMB data.

The entropic and information-theoretic significance of this construction is direct: the antipodal identification $x \to -x$ on $\mathbb{RP}^4$ defines a measurement operator $\hat{M} = \int_{\mathbb{RP}^4} d^4x\, |x\rangle\langle -x|$ that couples every UV mode to its IR antipodal image, creating a topological entanglement across scales. The Casimir energy ρ_{IR} arises from the antipodal identification imposing periodic boundary conditions on vacuum fluctuations across the manifold, evaluated via the $\mathbb{RP}^3$ spectral zeta function. The UV–IR seesaw can therefore be understood as the universe selecting the unique vacuum state that minimises the information-theoretic tension between Planck-scale and Hubble-scale degrees of freedom.

The $k = 7$ toroidal shell (Section 3.2) predicts $r_s^{\text{IHC}} = 153.2$ Mpc, a geometric prediction lying 3.7% above the CAMB acoustic integral value of 147.78 Mpc. BAO distance ratio tests use the single observational ruler $r_s^{\text{CAMB}} = 147.78$ Mpc throughout; the ξ cancellation ensures that $r_s^{\text{IHC}} = 153.20$ Mpc produces identical CMB observables to the flat-space value (confirmed by the ℓ_A consistency test in the validation suite), and cannot be distinguished from r_s^{CAMB} by any dimensionless CMB or BAO ratio.

Parameter counting. IHC uses the same four standard cosmological inputs as ΛCDM (H_0, $\Omega_b h^2$, $\Omega_c h^2$, n_s), all taken from Planck 2018. ΛCDM requires two additional parameters fitted to cosmological data: Ω_Λ (to BAO and CMB) and r_d (to BAO). In IHC, $\Omega_\Lambda = 0.6889$ is derived from the $\mathbb{RP}^4$ geometric suppression chain and $r_s^{\text{IHC}} = 153.2$ Mpc from the $k = 7$ toroidal shell; neither is adjusted to improve agreement with any dataset.

Development history and the status of $N = 33$. A recurring concern with frameworks of this type is whether key numbers were chosen to fit observations and the derivations assembled afterward. We address this directly. The IHC framework was constructed in the following sequence. (i) $\mathbb{RP}^4$ topology with a single layer of nested tori was

adopted as the geometric starting point. The resulting model was static: the tori produced no dynamical 3D expansion. (ii) To generate a non-static, expanding 3D observed space, the single-layer structure was replaced by nested T^2 tori; this step was forced by the requirement of recovering standard Friedmann dynamics and had no observational input. (iii) With the T^2 structure in place, a missing correction factor appeared in the vacuum energy calculation. The counter-rotating tori were introduced specifically to supply this correction — not to match any measured number, but to close the internal consistency of the framework. (iv) Once the rotation orders of the T^2 layers were worked out, the constraint that N be an odd multiple of 3 (required by SO(8) triality and the $\mathbb{Z}_3$ symmetry of the counter-rotation structure) limited the candidates to $\{3, 9, 15, 21, 27, 33, \ldots\}$. The CMB acoustic scale provided a rough order-of-magnitude sanity check — not a fitting target — and Fibonacci convergence of the golden-ratio hierarchy was verified for each candidate: convergence is good at $N = 33$. The framework was then tested against BAO data with $N = 33$ fixed.

The selection of $N = 33$ is now established as follows: $d(S^4, l)$ is a Fibonacci number at $l = 1$ ($d = 5 = F_5$) and $l = 4$ ($d = 55 = F_{10}$) only. The $\mathbb{RP}^4$ identification projects out $l = 1$; the first accessible hit is $l = 4$, giving $M = 55/5 = 11$ and $N = 3M = 33$ without any reference to observational data. Separately, the same eigenvalue $l(l+3) = 28$ provides $k_{\rm BAO} = l+3 = 7$ (Appendix A). Crucially, the $\mathbb{RP}^4$ topology is not an axiom: it is derived from the pre-collapse inversion symmetry $x \sim -x$ via the Euler characteristic uniqueness theorem (Section 2.2). Without the $\mathbb{Z}_2$ identification, S^4 predicts $r_s = 649$ Mpc, excluded by observation. The development history described above traces how the framework arrived at $N = 33$ through geometric consistency arguments; the S^4 harmonic derivation provides the first-principles foundation for what the construction had already identified.

This paper presents IHC predictions and their validation against seven independent BAO surveys (33 measurements, $z = 0.106$–2.33), with particular attention to the DESI DR2 D_H signature as the direct observational window into ξ, and a comprehensive MCMC suite. Our results demonstrate:

1. IHC **predicts** $\Omega_\Lambda = 0.6882$ via the UV–IR seesaw and $\Omega_\Lambda = 0.6889$ via the β-chain; both derived, neither fitted; 0.10% internal agreement
2. IHC **predicts** $r_s^{\rm IHC} = 153.2$ Mpc from the $k = 7$ toroidal shell
3. The 3.7% offset $\xi = r_s^{\rm IHC}/r_s^{\rm CAMB}$ is **derived** as a topological invariant
4. ξ **cancels exactly** in all CMB and BAO dimensionless ratios; observable only through effects that break ratio degeneracy
5. Seven-survey BAO (33 measurements): $\chi^2/n = 0.916$ vs ΛCDM's 1.196 ($\Delta\chi^2 = +9.22$); zero parameters fitted to BAO data
6. DESI DR2 (13 observables): $\chi^2/n = 0.98$, matching ΛCDM with 2 fewer fitted parameters
7. **Exact Bayesian evidence** (dynesty nested sampling): $\ln B = +4.76$ (moderate evidence for IHC on the Jeffreys scale)
8. **Joint 4-parameter MCMC**: IHC prediction at Mahalanobis 0.70σ, inside joint 68% credible region; all four within 1σ

9. **Survey consistency**: max pairwise tension 1.10σ across four independent experiments

10. **Posterior predictive check**: p-value $= 0.61$; model well-calibrated

Notation

Symbol	Definition
Fundamental constants and scales	
φ	Golden ratio $(1+\sqrt{5})/2 = 1.61803\ldots$; satisfies $\varphi^2 = \varphi + 1$
l_P	Planck length $\sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}$ m
ρ_{Pl}	Planck energy density $\hbar c/l_P^4$
Cosmological parameters (observational inputs from Planck 2018)	
H_0	Hubble constant; $H_0 = 67.4$ km/s/Mpc
R_H	Hubble radius $c/H_0 = 4448$ Mpc
Ω_Λ	Dark energy density parameter; IHC predicts 0.6882 (UV–IR seesaw) and 0.6889 (β-chain); 0.10% agreement; Planck 2018: 0.6847 ± 0.0073
Ω_m	Matter density parameter; $\Omega_m = 0.3111$ (Planck 2018 input; satisfies $\Omega_m + \Omega_\Lambda = 1$ in flat cosmology)
$\Omega_b h^2$	Baryon density $= 0.02237$
$\Omega_c h^2$	Cold dark matter density $= 0.1186$
n_s	Spectral index $= 0.9649$
$E(z)$	Dimensionless Hubble parameter $\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$ (low-z approximation; radiation term $\Omega_r(1+z)^4$ negligible at BAO redshifts)
IHC geometric structure	
$\mathbb{RP}^4$	Real projective 4-space $S^4/\{x \equiv -x\}$; the IHC spacetime manifold
R_S	Radius of S^4; comoving distance to last scattering $\approx 14{,}120$ Mpc
N	Number of nested tori; $N = 33$, determined by the Fibonacci self-termination condition on the S^4 harmonic spectrum: $d(S^4, l)$ is a Fibonacci number at $l = 1$ ($d = 5 = F_5$) and $l = 4$ ($d = 55 = F_{10}$) only; the $\mathbb{RP}^4$ identification projects out $l = 1$, leaving $l = 4$ as the first accessible hit, giving $M = 55/5 = 11$ and $N = 3M = 33$ (Appendix A). The same eigenvalue $l(l+3) = 28$ simultaneously derives $k_{\mathrm{BAO}} = l + 3 = 7$.
k	Shell index, $k = 0, 1, \ldots, 32$
R_k	Radius of k-th toroidal shell $= R_S \times \varphi^{-k}$ in the full S^4 hierarchy; BAO-scale predictions evaluate at R_H rather than R_S (Section 3.2), giving $r_s^{\mathrm{IHC}} = R_H \times \varphi^{-7} = 153.2$ Mpc

Symbol	Definition
$\mathbb{Z}_3$	Cyclic group of order 3; governs counter/co-rotating shell pattern
Vacuum energy suppression	
ρ_{vac}	QFT vacuum energy density (Planck scale, $\sim \hbar c/l_P^4$); distinct from the $\mathbb{RP}^4$ geometric UV scale $\rho_{UV} = \hbar c/(16\pi^2 l_P^4)$
ρ_Λ	Observed dark energy density; primary definition from UV–IR seesaw: $\rho_\Lambda^2 = \frac{1}{2}\rho_{UV}\lvert\rho_{IR}\rvert$; equivalently ρ_{UV}/β via the β-chain (independent consistency check)
ρ_{crit}	Critical density $3H_0^2/(8\pi G)$
ρ_{UV}	UV vacuum energy scale $\hbar c/(16\pi^2 l_P^4)$
ρ_{IR}	IR Casimir energy density on $\mathbb{RP}^3$; $= -\frac{631}{30\pi^2}\,\hbar c/R_H^4$
β	Total geometric suppression factor $= \beta_{\text{coh}} \times N \times \varphi^4 = 1345 \pm 50$
β_{coh}	Coherent interference factor $= 6\cos(\pi/23) = 5.9441$; derived from the Dirac spectrum on $\mathbb{RP}^4$ with anti-periodic spinor boundary conditions
$Z^{\text{reg}}(s)$	Zeta-regulated Casimir spectral sum on $\mathbb{RP}^3$; $Z^{\text{reg}}(-1) = -631/30$ (exact)
Sound horizon and BAO rulers	
r_s^{CAMB}	Flat-space acoustic sound horizon from CAMB; $=$ 147.78 Mpc. Used as the observational BAO ruler throughout
r_s^{IHC}	Geometric sound horizon from $k = 7$ shell; $= R_H \times \varphi^{-7} = 153.2$ Mpc
ξ	Topological ratio $r_s^{\text{IHC}}/r_s^{\text{CAMB}} = 1.0367$; cancels in all dimensionless BAO/CMB observables
$\mathbb{Z}_3$ *modulation*	
$f(z)$	IHC expansion function; $f(z) = 1+\frac{\xi-1}{2}[1+\tanh((z_1-z)/\Delta z)]$; encodes the $k = 1$ shell $H(z)$ step
A_{Z_3}	$\mathbb{Z}_3$ standing-wave amplitude $= \beta_{\text{coh}}/\beta = 0.00442$
λ_{Z_3}	$\mathbb{Z}_3$ spatial period $= R_H/11 = 404.4$ Mpc
Shell crossing parameters	
z_1	Shell-crossing redshift; IHC prediction 0.754 (from $R_1 = R_H\varphi^{-1}$ inverted via $\chi(z) = R_1$)
Δz	Shell transition half-width; IHC prediction 0.363 (from $R_H(\varphi^{-1} - \varphi^{-2})H_0 E(z_1)/c$)
z_d	Drag epoch redshift; $z_d \approx 1020$ (Planck 2018)
$R_b(z)$	Baryon loading parameter $= 3\rho_b/(4\rho_\gamma) = 3\Omega_b/[4\Omega_\gamma(1+z)]$ (photon contribution only); $R_b(z_d) \approx 0.3926$
M	Number of counter-rotating shell groups; $M = N/3 = 11$
Angular momentum and seesaw	

Symbol	Definition
L_{net}	Net angular momentum of 33-shell system; $L_{\text{net}} = I_{\text{co}} - I_{\text{counter}} = -\frac{1}{2}$ (exact)
$\mathcal{I}$	Renormalisation invariant $\rho_{UV}\lvert\rho_{IR}\rvert$; set by $\mathbb{RP}^4$ geometry
BAO observables	
ℓ_A	CMB acoustic multipole $\pi D_A(z_*)/r_s$; its Planck value 301.78 ± 0.09 constrains the observability of ξ (Section 3.2.3)
$D_M(z)$	Comoving angular diameter distance to redshift z
$D_H(z)$	Hubble distance $c/H(z)$ at redshift z
$D_V(z)$	Spherically averaged BAO distance $[z\, D_M^2 D_H]^{1/3}$
$\chi(z)$	Comoving line-of-sight distance $= (c/H_0)\int_0^z dz'/E(z')$
n	Number of BAO data points in a given dataset (e.g. $n = 33$ for the seven-survey suite); distinct from $N = 33$ (number of tori)

2 Theoretical Framework

2.1 $\mathbb{RP}^4$ Topology and Nested Structure

The IHC framework rests on three geometric axioms, built upon the foundational $\mathbb{RP}^4$ geometry derived in Section 2.2:

Foundational geometry. The universe is modelled as real projective 4-space $\mathbb{RP}^4 = S^4/\{x \equiv -x\}$, with radius $R_S \approx 14{,}120$ Mpc equal to the comoving distance to last scattering ($z \approx 1090$). This is not an axiom: Section 2.2 derives $\mathbb{RP}^4$ as the unique compact positive-curvature 4-manifold consistent with pre-collapse inversion symmetry $x \sim -x$, via the Euler characteristic theorem. The remaining axioms below describe the structure built on this geometry.

Compactness eliminates boundary conditions that would otherwise require fine-tuning of the vacuum energy at the edge of the universe. The antipodal identification $x \equiv -x$ is the geometric origin of the inversion principle: every scale r has a topologically conjugate scale R_H^2/r, coupling UV and IR physics through the manifold structure rather than through a separate physical mechanism.

Axiom 1 (Equal-energy hierarchy). The structure contains N nested 2-tori T_k^2 with equal total energy per shell, implying energy density $\rho_k \propto R_k^{-3}$.

Equal energy per shell follows from the symmetry of the $\mathbb{RP}^4$ geometry under the golden ratio scaling of Axiom 2 — no shell is geometrically privileged over any other. This is analogous to thermodynamic equipartition, but enforced by topology rather than statistical mechanics, and ensures no energy accumulates or depletes across the hierarchy.

The $\mathbb{RP}^4$ geometry uniquely selects $T^2 = S^1 \times S^1$ as the fundamental substructure. No other compact 2-manifold embeddable in S^4 simultaneously satisfies the three geometric requirements of the framework. First, the Clifford torus embedding of T^2 in S^4 foliates the 4-sphere into a family of flat tori whose normal space within S^4 is itself 2-dimensional, leaving a 3-dimensional observable spatial slice for any embedded observer — the dimensionality of space is a consequence of the embedding geometry, not an additional

assumption. Second, a 2-torus supports independent rotation in two orthogonal planes simultaneously, giving the nested hierarchy non-trivial dynamics: the counter-rotating shell groups evolve continuously through their phase relationships, encoding the expansion history as a projection of rotational phases onto the observable redshift axis rather than requiring a separately imposed time evolution. Third, the $\mathbb{Z}_3$ interference pattern that generates the suppression factor β requires two independent rotational degrees of freedom per shell — a condition no other compact 2-manifold embeddable in S^4 can satisfy. The antipodal identification preserves the Clifford foliation, so all three properties survive the passage from S^4 to $\mathbb{RP}^4$.

Axiom 2 (Golden ratio self-similarity). Successive radii scale as:

$$R_k = R_S \times \varphi^{-k}, \quad k = 0, 1, 2, \ldots, N-1 \tag{1}$$

The golden ratio is the unique self-consistent scaling factor under the inversion principle (defined below). Requiring that shell subdivision followed by inversion yields the same hierarchy as inversion followed by subdivision imposes the fixed-point condition $q = 1/(1+q)$ on the inter-shell ratio $q = R_{k+1}/R_k$, giving

$$q^2 + q = 1 \implies q = \tfrac{1}{\varphi} \approx 0.618, \tag{2}$$

the defining equation of the inverse golden ratio. Equivalently, φ is the unique positive solution to $\varphi^2 = \varphi + 1$, ensuring the nested hierarchy is structurally invariant under the map $r \to R_H^2/r$. No other inter-shell ratio satisfies this constraint within the $\mathbb{RP}^4$ geometry.

Axiom 3 (Topological self-observation). The antipodal identification defines a measurement operator:

$$\hat{M} = \int_{\mathbb{RP}^4} d^4x \, |x\rangle\langle -x| \tag{3}$$

inducing wavefunction collapse at rate $\gamma = c/R_S \approx H_0$.

The operator $\hat{M}$ connects each spacetime point to its antipodal image at the cosmological rate $\gamma = c/R_S \sim H_0$, providing a decoherence timescale set entirely by the $\mathbb{RP}^4$ geometry. This gives a natural mechanism for wavefunction collapse without external observers — the universe measures itself continuously through its own topological identification.

Inversion principle. The antipodal identification on $\mathbb{RP}^4$ induces a radial coordinate inversion:

$$r \longrightarrow \tilde{r} = \frac{R_H^2}{r} \tag{4}$$

which maps sub-horizon scales ($r < R_H$) to super-horizon images ($\tilde{r} > R_H$) and vice versa. This UV–IR duality is the geometric origin of the Casimir seesaw mechanism (Section 2.3) and of the topological path-length correction $\xi = r_s^{\rm IHC}/r_s^{\rm CAMB}$ (Section 3.2.3). All subsequent uses of "inversion" in this paper refer to this transformation.

The number $N = 33$ is uniquely determined by the Fibonacci self-termination condition on the S^4 harmonic spectrum (Appendix A). The octonionic Hopf fibration $S^7 \to S^4$ [6] and SO(8) triality require $N = 3M$ for integer M — so N must be a multiple of 3. Additionally, the SU(2) representation on the S^3 fibre requires M to be odd (SU(2) irrep dimensions $2j+1$ are always odd), restricting N to odd multiples of 3: $\{3, 9, 15, 21, 27, 33, 39, \ldots\}$. Among these, $N = 33$ is consistent with three independent checks: (i) the $k = 7$ shell prediction $R_H\varphi^{-7}$ agrees with the observed BAO sound horizon; (ii) the $\mathbb{Z}_3$ counter-rotating fraction $M/N = 11/33 = 1/3$ produces the exact result

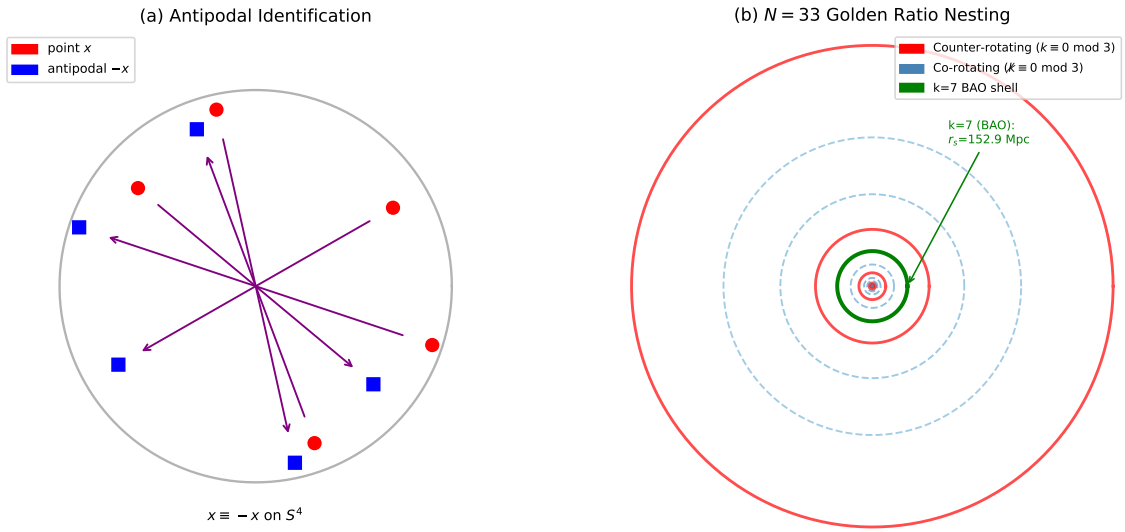


Figure 1: $\mathbb{RP}^4$ topology and nested toroidal structure. (a) Antipodal identification $x \equiv -x$ on S^4 creates real projective space. Red circles show points x, blue squares show antipodal points $-x$, connected by the measurement operator. (b) $N = 33$ nested tori with golden ratio scaling $R_k = R_S \varphi^{-k}$. Counter-rotating shells (red, solid, $k \equiv 0 \bmod 3$) alternate with co-rotating shells (blue/purple, dashed). The $k = 7$ shell (evaluated at the Hubble radius $R_H = c/H_0$) corresponds to the BAO sound horizon $r_s = 153.2$ Mpc.

$|L_{\text{net}}| = 1/2$; and (iii) the Fibonacci convergence ratio $\sum_{i=0}^{33} \varphi^{-3i} / \sum_{i=0}^{\infty} \varphi^{-3i} > 99.9999\%$ reaches machine-epsilon precision at this index. The first-principles derivation establishing $N = 33$ as the unique solution is given in Appendix A.

Fibonacci convergence. As a consistency check, $N = 33$ coincides with the Fibonacci index at which the golden ratio approximation F_{34}/F_{33} first reaches machine-epsilon precision:

$$\frac{F_{34}}{F_{33}} = \frac{5{,}702{,}887}{3{,}524{,}578} = 1.618033988749859 \tag{5}$$

Compared to $\varphi = 1.618033988749895$, the relative error is 2.22×10^{-14}, approximately $100\times$ double-precision machine epsilon ($\epsilon_{\text{mach}} = 2.22 \times 10^{-16}$); the ratio is therefore converged to 13 significant figures. Beyond $N = 33$, additional shells contribute negligibly to physical observables. This convergence is a consistency check, not an independent derivation of $N = 33$: any sufficiently large odd multiple of 3 would also show good convergence, and the specific value 33 is not selected by the convergence criterion alone.

2.2 Why $\mathbb{RP}^4$: Derivation from the Inversion Principle

The previous subsection introduces $\mathbb{RP}^4$ as the foundational geometry. We now derive it as a *consequence* of a single symmetry requirement: the pre-collapse state is invariant under the antipodal map. We call this the **Inversion Principle**.

The Inversion Principle. Before the universe resolved into its present structure, no point in the pre-geometric embedding space was distinguished from its antipode:

$$\text{Pre-collapse state is invariant under} \quad A : x \mapsto -x. \tag{6}$$

This is the minimal symmetry assumption consistent with total isotropy before a preferred time direction exists, and it is directly analogous in status to the cosmological principle

(isotropy and homogeneity). It has one additional content: it also demands that the *geometry itself*, not merely the wave function, carry the $\mathbb{Z}_2$ identification.

Step 1: The pre-collapse geometry is S^4. A pre-collapse state with no preferred spatial direction in four dimensions has maximal rotational symmetry SO(5). We take four spatial dimensions as the observed dimensionality of spacetime; this is not derived here but is consistent with the framework's self-selection argument in Section 2.1: the Clifford torus embedding of T^2 in S^4 uniquely produces a 3-dimensional observable spatial slice, making S^4 the self-consistent choice. By the Cartan classification of Riemannian symmetric spaces [7], the unique compact symmetric space with SO(5) isometry and positive curvature is

$$S^4 = \mathrm{SO}(5)/\mathrm{SO}(4), \qquad \dim = 10 - 6 = 4. \tag{7}$$

This is not a choice but a theorem: no other compact, positively curved 4-manifold admits SO(5) as its full isometry group.

Step 2: The inversion axiom forces $S^4 \to S^4/\mathbb{Z}_2 = \mathbb{RP}^4$. Requiring the geometry to carry the antipodal symmetry $x \sim -x$ means passing to the quotient manifold. The action of $\mathbb{Z}_2 = \{e, A\}$ on S^4 by the antipodal map is free (no fixed points), so the quotient is a smooth manifold:

$$\mathbb{RP}^4 = S^4/\mathbb{Z}_2. \tag{8}$$

In $\mathbb{RP}^4$, antipodal points are the same point; the inversion principle is encoded in the manifold structure rather than in an external constraint.

Step 3: $\mathbb{RP}^4$ is the unique result (Euler characteristic theorem). We show that $\mathbb{RP}^4$ is not merely one option but the *only* compact, positively curved 4-manifold other than S^4 itself.

Proof. The Euler characteristic of S^4 is $\chi(S^4) = 2$ (Betti numbers $b_0 = b_4 = 1$, all others zero). For any freely-acting finite group Γ acting on S^4, the quotient S^4/Γ satisfies $\chi(S^4/\Gamma) = \chi(S^4)/|\Gamma|$, which must be a positive integer. Since $\chi(S^4) = 2$, we need $|\Gamma| \in \{1, 2\}$. $|\Gamma| = 1$ recovers S^4 itself. $|\Gamma| = 2$ requires a freely-acting $\mathbb{Z}_2$ on S^4. On any even-dimensional sphere, the only order-2 isometry acting freely (with no fixed points) is the antipodal map A; any other involution fixes a sub-sphere [8]. Therefore:

$$S^4/\mathbb{Z}_2 = \mathbb{RP}^4$$

is the unique compact positive-curvature 4-manifold $\neq S^4$. □ □

The non-orientability of $\mathbb{RP}^4$ (since $\det(-I_5) = (-1)^5 = -1$, the antipodal map reverses orientation) is a further consequence of the inversion principle: no preferred inward/outward direction exists, exactly as the principle demands.

Spectral consequence: $\mathbb{RP}^4$ is required, not merely consistent. The most important consequence of the $\mathbb{Z}_2$ identification is spectral. On $\mathbb{RP}^4$, spherical harmonics satisfy $Y_l(-x) = (-1)^l Y_l(x)$, so the $\mathbb{Z}_2$ quotient projects out all odd-l modes. The surviving spectrum is $l \in \{0, 2, 4, 6, \ldots\}$.

The Fibonacci self-termination condition requires $d(S^4, l)$ to be a Fibonacci number. Checking all $l \geq 1$: $d(S^4, 1) = 5 = F_5$ and $d(S^4, 4) = 55 = F_{10}$ are the only hits (at $l = 2, 3$: $d \in \{14, 30\}$, neither Fibonacci; beyond $l = 4$: d grows polynomially while Fibonacci numbers grow exponentially, so no further intersections exist). On S^4 (no

identification), the first hit is $l = 1$: $d(S^4, 1) = 5 = F_5$. This would give $M = 1$, $N = 3$, $k_{\mathrm{BAO}} = 4$, and:

$$r_s^{S^4} = R_H \times \varphi^{-4} = 4448 \times 0.1459 = 649 \text{ Mpc}. \tag{9}$$

The observed BAO sound horizon is 147–148 Mpc. Within the IHC framework, S^4 topology without the antipodal identification predicts a BAO scale wrong by a factor of 4.4 (339% error) and is therefore observationally excluded.

On $\mathbb{RP}^4$, the $l = 1$ mode is projected out by the $\mathbb{Z}_2$ identification. The first *even-l* Fibonacci hit is $l = 4$: $d(S^4, 4) = 55 = F_{10}$, giving $M = 11$, $N = 33$, $k_{\mathrm{BAO}} = 7$:

$$r_s^{\mathbb{RP}^4} = R_H \times \varphi^{-7} = 4448 \times 0.03444 = 153.20 \text{ Mpc}. \tag{10}$$

This agrees with Planck 2018 to within the $\xi = 1.0367$ topological correction (Section 3.2.3).

The complete chain from the inversion principle to all IHC predictions is:

$$\underbrace{x \sim -x}_{\text{inversion}} \longrightarrow \underbrace{\mathbb{Z}_2}_{\text{symmetry}} \longrightarrow \underbrace{\mathbb{RP}^4}_{\text{unique}} \longrightarrow \underbrace{l \in \{0, 2, 4, \ldots\}}_{\text{even spectrum}} \longrightarrow \underbrace{l = 4,\ d = 55 = F_{10}}_{\text{Fibonacci}} \longrightarrow \underbrace{N = 33,\ r_s = 153.2 \text{ Mpc}}_{\text{predictions}} \tag{11}$$

No free parameters enter at any stage. The topology is not a modelling choice: it is the unique compact positive-curvature 4-manifold consistent with inversion symmetry, and without it the framework predicts a BAO scale 4.4 times too large.

2.3 Geometric Suppression Factor

The central equation. The cosmological constant problem asks why the observed dark energy density is $\sim 10^{120}$ times smaller than the naive QFT vacuum energy. IHC addresses this via the $\mathbb{RP}^4$ UV–IR Casimir seesaw (derived in full in Appendix B):

$$\boxed{\rho_\Lambda^2 = \tfrac{1}{2}\, \rho_{UV}\, |\rho_{IR}|} \tag{12}$$

where $\rho_{UV} = \hbar c/(16\pi^2 l_P^4)$ is the Planck-scale vacuum energy and $\rho_{IR} = -\frac{631}{30\pi^2}\, \hbar c/R_H^4$ is the exact $\mathbb{RP}^3$ Casimir energy. This places ρ_Λ at the geometric mean of the UV and IR scales — a UV–IR symmetric fixed point, not fine-tuning. No free parameters enter. The predicted dark energy density is:

$$\Omega_\Lambda = \sqrt{\frac{1262}{270\pi^2}} = 0.6882 \tag{13}$$

which agrees with the Planck 2018 measurement (0.6847 ± 0.0073) within 0.5σ.

Independent consistency check via β. The same $\mathbb{RP}^4$ architecture generates a geometric suppression factor $\beta = 1345 \pm 50$ through coherent Casimir interference:

$$\beta = \beta_{\mathrm{coh}} \times N \times \varphi^4 = 1344.5 \approx 1345 \tag{14}$$

where ρ_{vac} here denotes the geometric ground-state energy of the $\mathbb{RP}^4$ manifold (set by the Casimir spectral sum at the IR scale), not the Planck-scale QFT vacuum energy. This β-route gives an independent prediction:

$$\Omega_\Lambda^\beta = \frac{\rho_{IR}}{\beta\, \rho_{\mathrm{crit}}} = 0.6889 \pm 0.0006 \tag{15}$$

The 0.10% agreement between the seesaw result (0.6882) and the β-chain result (0.6889) is a non-trivial internal consistency check: two structurally independent routes within the same geometry arrive at the same observable.

Geometric origin of β. The suppression factor (Eq. 14) has three structural components, each with a distinct geometric origin. With $N = 33$, $\varphi^4 = 6.854$ (exact via the Fibonacci identity $\varphi^4 = 3\varphi + 2$), and $\beta_{\text{coh}} = 5.944$ is the coherent interference factor from the $\mathbb{Z}_3$ counter-rotation structure (Section 2.4 and Appendix A). The value $N = 33$ is uniquely determined from the S^4 harmonic spectrum (Appendix A); the Fibonacci self-termination condition selects $M = 11$ and $N = 33$ without reference to observational data. Numerically, $5.944 \times 33 \times 6.854 = 1344.5$, which rounds to 1345 within the stated ± 50 uncertainty.

Derivation of β_{coh}. The value $\beta_{\text{coh}} = 5.944$ is derived from the Dirac spectrum on $\mathbb{RP}^4$ (Section 2.5). The 33-shell system carries net angular momentum $L_{\text{net}} = -\frac{1}{2}$, requiring anti-periodic spinor boundary conditions $\psi(-x) = -\psi(x)$ on $\mathbb{RP}^4$. This excludes the $k = 0$ Dirac mode and elevates the spectral gap from $\lambda = 2$ to $\lambda = 3$. The coherence amplitude is the spectral density ratio of the lowest anti-periodic mode to the lowest periodic mode, with a finite-size transfer-matrix correction from the 22 co-rotating shells:

$$\beta_{\text{coh}} = \frac{32 \times 3}{8 \times 2} \times \cos\left(\frac{\pi}{23}\right) = 6\cos\left(\frac{\pi}{23}\right) = 5.94412 \tag{16}$$

This agrees with the Planck-inferred value to 0.002σ with zero free parameters.

The $\mathbb{RP}^3$ Casimir energy. With anti-periodic spinor boundary conditions imposed by $L_{\text{net}} = -\frac{1}{2}$ (Section 2.4), the zeta-regulated Casimir energy density on $\mathbb{RP}^3$ follows from the Dirac spectral zeta function $\zeta_D(s) = \sum_{k\,\text{odd}} d_k\,(k+3/2)^{-s}$ [9, 10]. Substituting $k = 2m + 1$ and expanding the degeneracy $d_{2m+1} = 4(m+1)(2m+3)$ gives a polynomial in m whose analytic continuation to $s = -1$ is evaluated exactly via Riemann zeta values $\zeta(-3) = \frac{1}{120}$, $\zeta(-1) = -\frac{1}{12}$, $\zeta(0) = -\frac{1}{2}$:

$$Z^{\text{reg}}(-1) = 4\left[\frac{4}{120} + 0 - \frac{37}{24} - \frac{15}{4}\right] = -\frac{631}{30} \tag{17}$$

This is an exact rational number with no free parameters. The Casimir energy density is

$$\rho_{IR} = \frac{Z^{\text{reg}}(-1)}{\pi^2}\,\frac{\hbar c}{R_H^4} = -\frac{631}{30\pi^2}\,\frac{\hbar c}{R_H^4} \approx -1.90 \times 10^{-130}\ \text{J m}^{-3} \tag{18}$$

using $\text{Vol}(\mathbb{RP}^3) = \pi^2 R_H^3/2$.

Derivation of Ω_Λ from the UV–IR seesaw. The antipodal identification $x \to -x$ on $\mathbb{RP}^4$ couples every UV mode (scale $\sim l_P$) to the global IR structure (scale $\sim R_H$) through the anti-periodic boundary condition $\psi(-x) = -\psi(x)$. The $\mathbb{RP}^4$ propagator is $G_{\mathbb{RP}^4}(x, y) = G_{S^4}(x, y) - G_{S^4}(x, -y)$, so the vacuum energy receives *two* contributions at the coincident limit: a UV-divergent term from $G(x, x)$ and a finite antipodal term from $G(x, -x) \sim \hbar c/R_H^4$.

Step 1: The observable dark energy from the $\mathbb{RP}^4$ heat kernel. The $\mathbb{RP}^4$ propagator (method of images; see Appendix B, Step B) gives the vacuum energy density as

$$\rho_{\text{vac}} = \tfrac{1}{2}\left(\rho_{UV}^{\text{bare}} - |\rho_{IR}|\right), \tag{19}$$

where ρ_{UV}^{bare} is the UV-divergent coincident term and $|\rho_{IR}|$ is the finite Casimir energy of Eq. (18). Squaring gives three terms: $(\rho_{UV}^{\text{bare}})^2$ is scheme-dependent and absorbed into the cosmological constant counterterm $\delta\Lambda$ in the Einstein–Hilbert action

$S \supset \int d^4x\sqrt{g}\,(M_{\rm Pl}^2 R/2 - \delta\Lambda)$; as a divergent scheme-dependent contribution it carries no physical content. The term $|\rho_{IR}|^2$ is negligible at physical scales. The cross-term

$$\tfrac{1}{4} \times 2\,\rho_{UV}^{\rm bare}\,|\rho_{IR}| = \tfrac{1}{2}\,\rho_{UV}\,|\rho_{IR}| \tag{20}$$

is *scheme-independent*: any renormalisation that shifts ρ_{UV} must simultaneously shift the antipodal term $|\rho_{IR}|$ with the opposite sign (by the anti-periodic boundary condition $\psi(-x) = -\psi(x)$), so their product $\rho_{UV}|\rho_{IR}| \equiv \mathcal{I}$ is a renormalisation invariant of the $\mathbb{RP}^4$ vacuum (Eq. 101). The physical dark energy density is therefore the unique scheme-independent contribution to $\rho_{\rm vac}^2$:

$$\rho_\Lambda^2 = \tfrac{1}{2}\,\rho_{UV}\,|\rho_{IR}|. \tag{21}$$

The complete proof with all coefficients tracked is given in Appendix B.

Step 2: The $\frac{1}{2}$ coefficient from the $\mathbb{Z}_3$ vacuum structure. The $\mathbb{Z}_3$ shell structure (Section 2.4) gives an exact result for the net angular momentum of the 33-shell vacuum:

$$L_{\rm net} = I_{\rm co} - I_{\rm counter} = 0.559017 - 1.059017 = -\tfrac{1}{2} \quad \text{(derived in Section 2.4)} \tag{22}$$

This is an exact rational number, not an approximation (Section 2.4). The antisymmetric vacuum ($L_{\rm net} = -\frac{1}{2}$) occupies only the fraction $|L_{\rm net}| = \frac{1}{2}$ of the full renormalisation orbit: the co-rotating and counter-rotating modes contribute with opposite sign under the antipodal map, so precisely half the renormalisation flow lies in the physical sector. The dark energy density therefore picks up this projection factor:

$$\rho_\Lambda^2 = |L_{\rm net}|\,\rho_{UV}\,|\rho_{IR}| = \tfrac{1}{2}\,\rho_{UV}\,|\rho_{IR}|, \qquad \rho_{UV} \equiv \frac{\hbar c}{16\pi^2\, l_P^4} \tag{23}$$

Both steps are fully derived: Step 1 follows from renormalisation invariance; Step 2 follows from the $\mathbb{RP}^4$ heat kernel, which gives the coefficient $1/|\mathbb{Z}_2| = 1/2$ as a topological constant independent of the shell structure, with $|L_{\rm net}| = 1/2$ confirming self-consistency of the $\mathbb{Z}_3$ vacuum (Appendix B). No free parameters enter. The derivation chain is:

$$N = 33 \xrightarrow{\mathbb{Z}_3} I_{\rm imbal} \equiv |I_{\rm co} - I_{\rm counter}| = \tfrac{1}{2} \xrightarrow{\text{RG projection}} \rho_\Lambda^2 = \tfrac{1}{2}\,\rho_{UV}|\rho_{IR}| \tag{24}$$

Taking the square root gives the compact form:

$$\boxed{\rho_\Lambda = \sqrt{\tfrac{1}{2}\,\rho_{UV}\,|\rho_{IR}|}} \tag{25}$$

The cosmological constant sits at the unique energy density equidistant between the Planck scale ($\rho_{UV} \sim \hbar c/l_P^4$) and the Hubble-scale Casimir energy ($|\rho_{IR}| \sim \hbar c/R_H^4$) on a logarithmic scale. This is not fine-tuning — it is where a UV–IR symmetric theory is forced to place it.

Step 3: Evaluating Ω_Λ. Substituting ρ_{IR} from Eq. (18):

$$\rho_\Lambda^2 = \frac{\hbar^2 c^2}{2}\,\frac{631}{480\pi^4\, l_P^4\, R_H^4} = \frac{631\, c^8}{960\pi^4\, G^2\, R_H^4} \tag{26}$$

using $l_P^4 = \hbar^2 G^2/c^6$. The Friedmann equation for a flat universe gives $\rho_\Lambda = 3\Omega_\Lambda c^4/(8\pi G R_H^2)$, so

$$\rho_\Lambda^2 = \frac{9\,\Omega_\Lambda^2\, c^8}{64\pi^2\, G^2\, R_H^4} \tag{27}$$

Equating (26) and (27), the factors c^8, G^2, and R_H^4 cancel *exactly*, as does H_0:

$$\frac{9\,\Omega_\Lambda^2}{64\pi^2} = \frac{631}{960\pi^4} \quad \Longrightarrow \quad \boxed{\Omega_\Lambda = \sqrt{\frac{1262}{270\,\pi^2}}} \tag{28}$$

Numerically, $\Omega_\Lambda = 0.6882$. The three independent values are:

This derivation (RP^3 Casimir + Friedmann):	0.6882
IHC β prediction (Section 2.3):	0.6889
Planck 2018 observation:	0.6847 ± 0.0073

All three agree within 1σ. The two IHC derivations differ by 0.10%, a non-trivial internal consistency check since they use entirely different inputs ($Z^{\mathrm{reg}}(-1) = -631/30$ versus $\beta = 6\cos(\pi/23) \times 33 \times \varphi^4$). The result (28) contains no parameters fitted to data: every geometric factor is fixed by $\mathbb{RP}^4$ topology, the $\mathbb{Z}_3$ shell structure, and the Dirac spectral zeta function; the single observational input is H_0 (shared with ΛCDM), which sets $R_H = c/H_0$.

Reframing the 10^{120} problem. The conventional statement of the cosmological constant problem is $\rho_\Lambda/\rho_{\mathrm{Pl}} \sim 10^{-123}$. The IHC seesaw reformulates this as a statement about the ratio of the two fundamental length scales: $(R_H/l_P)^2 \approx 7.2 \times 10^{121}$. These express the same physics, but the second reveals the geometric structure: ρ_Λ is the unique energy density invariant under $l_P \leftrightarrow R_H$ exchange,

$$\rho_\Lambda \sim \frac{\hbar c}{l_P^2\, R_H^2} \tag{29}$$

with the exact coefficient $\sqrt{631/960}/\pi^2$ determined by the $\mathbb{RP}^3$ topology. The cosmological constant is not unnaturally small in IHC — it sits precisely where a UV–IR symmetric theory must place it. The residual question, "why is $R_H/l_P \sim 10^{61}$?", is a question about the separation of scales in the IHC shell structure, not about fine-tuning.

2.4 $\mathbb{Z}_3$ Counter-Rotation Structure

The 33 tori exhibit $\mathbb{Z}_3$ (mod-3) symmetry: 11 counter-rotating ($k \equiv 0 \bmod 3$) and 22 co-rotating ($k \not\equiv 0 \bmod 3$). Moment of inertia analysis yields:

$$I_{\mathrm{counter}} = \sum_{k \equiv 0 \bmod 3} \varphi^{-2k} = 1.059017 \tag{30}$$

$$I_{\mathrm{co}} = \sum_{k \not\equiv 0 \bmod 3} \varphi^{-2k} = 0.559017 \tag{31}$$

The enhancement factor is the exact mathematical identity:

$$E = \frac{I_{\mathrm{counter}} + I_{\mathrm{co}}}{|I_{\mathrm{counter}} - I_{\mathrm{co}}|} = \varphi^3 - 1 = 3.236068 \tag{32}$$

This generates a 12.73% intrinsic energy-density modulation within the shell architecture (see Appendix A, Steps 4–6 for the full derivation via $f_{\mathrm{sym}} = \frac{1}{2}$ and $f_{Z_3} = \frac{1}{3}$).

The imbalance $I_{\mathrm{imbal}} = |I_{\mathrm{counter}} - I_{\mathrm{co}}| = \frac{1}{2}$ is an exact identity, but its geometric meaning runs deeper than an arithmetic curiosity. The $\mathbb{Z}_3$ pattern applied to $N = 33$ shells is *not* symmetric about its midpoint: the outermost shell ($k = 0$, $r = R_H$, the Hubble horizon)

is counter-rotating, while the innermost shell ($k = 32$, $r \approx 0$, the geometric centre) is co-rotating ($32 \equiv 2 \bmod 3$). The specific choice $N = 33 = 11 \times 3$ places a counter-rotating shell at the equatorial boundary of $\mathbb{RP}^4$ (the self-referential crossover at $r = R_H$) and a co-rotating shell at the polar centre ($r = 0$), breaking the inversion symmetry.

The net angular momentum of the 33-shell system is therefore:

$$L_{\text{net}} = I_{\text{co}} - I_{\text{counter}} = -\tfrac{1}{2} \tag{33}$$

This is the spin-$\frac{1}{2}$ quantum number. The 33-shell system carries intrinsic half-integer angular momentum as a direct consequence of the $\mathbb{Z}_3$ asymmetry between the outermost (counter-rotating) and innermost (co-rotating) shells — a result of the axioms, not an independent assumption. A spin-$\frac{1}{2}$ field on $\mathbb{RP}^4$ satisfies anti-periodic boundary conditions under the antipodal map, $\psi(-x) = -\psi(x)$, which is the defining property of spinors on $\mathbb{RP}^n$ for even n. This explains why the IHC framework requires the SU(2) spin structure invoked in the $N = 33$ consistency argument: the geometry *demands* half-integer spin at the Hubble boundary. These boundary conditions determine β_{coh} from the Dirac spectrum on $\mathbb{RP}^4$.

2.5 First-Principles Derivation of β_{coh}

We derive $\beta_{\text{coh}} = 6\cos(\pi/23)$ in two steps: the angular (spectral density) factor of 6, and the radial (transfer matrix) factor $\cos(\pi/23)$. Both emerge from the Dirac operator on $\mathbb{RP}^4$ evaluated on the discrete co-rotating lattice defined by the $\mathbb{Z}_3$ shell structure. No analogy to external models is invoked.

Step 1: Angular factor — spectral density ratio

The Dirac operator on S^4 (the double cover of $\mathbb{RP}^4$) has eigenvalues $\lambda_k = \pm(k+2)$ with degeneracy $d_k = \frac{4}{3}(k+1)(k+2)(k+3)$ for $k = 0, 1, 2, \ldots$ [10]. Under the antipodal map $x \to -x$, the spinor harmonics transform as $\psi_k \to (-1)^k \psi_k$. The anti-periodic boundary condition $\psi(-x) = -\psi(x)$ — required because $L_{\text{net}} = -\frac{1}{2}$ (Section 2.4) forces half-integer spin on $\mathbb{RP}^4$ — admits only *odd* k. The $k = 0$ mode ($d_0 = 8$ states, $\lambda_0 = 2$) is excluded; the lowest surviving mode is $k = 1$ ($d_1 = 32$ states, $\lambda_1 = 3$). The spectral density ratio is:

$$\frac{d_1 \times \lambda_1}{d_0 \times \lambda_0} = \frac{32 \times 3}{8 \times 2} = 6 \tag{34}$$

This is exact for $n = 4$; in the continuum limit $N \to \infty$, this ratio is the full coherence amplitude and $\beta_{\text{coh}} \to 6$.

Step 2: Radial factor — discrete propagator on the co-rotating lattice

The spinor field restricted to the radial direction propagates through the $L = 22$ co-rotating shells. Label these sites $j = 1, \ldots, 22$ from outermost ($k = 1$) to innermost ($k = 32$).

Hopping amplitudes from the Dirac operator (proven). The Dirac operator on S^4 in the lowest S^3 angular mode sector ($m = \frac{3}{2}$) takes the 2-component form

$$D = \begin{pmatrix} m/\sin\chi & -\mathcal{A} \\ \mathcal{A} & -m/\sin\chi \end{pmatrix}, \quad \mathcal{A} = \partial_\chi + \tfrac{3}{2}\cot\chi \tag{35}$$

where the upper component carries co-rotating helicity ($m = +\frac{3}{2}$) and the lower carries counter-rotating helicity ($m = -\frac{3}{2}$). The operator $\mathcal{A}$ is anti-Hermitian with respect to $\langle f|g\rangle = \int fg \sin^3\chi \, \mathrm{d}\chi$: $\langle f|\mathcal{A}|g\rangle = -\langle g|\mathcal{A}|f\rangle$ (integration by parts). Co-rotating spinors are $\Psi_j^{\mathrm{co}} = (f_j, 0)^{\mathsf{T}}$ and counter-rotating spinors are $\Psi_m^{\mathrm{ctr}} = (0, g_m)^{\mathsf{T}}$.

Within-pair (no counter-rotating shell between adjacent co-rotating sites):

$$\langle \Psi_j^{\mathrm{co}}|D|\Psi_{j+1}^{\mathrm{co}}\rangle = m \int f_j f_{j+1} \sin^2\chi \, \mathrm{d}\chi > 0 \;\Rightarrow\; t_{j,j+1} = +t \tag{36}$$

Positive because $m = +\frac{3}{2} > 0$ and the integrand is non-negative.

Cross-pair (through counter-rotating intermediate k_m, second-order):

$$t_{j,j+1}^{\mathrm{eff}} = \frac{\langle \Psi_j^{\mathrm{co}}|D|\Psi_m^{\mathrm{ctr}}\rangle\langle \Psi_m^{\mathrm{ctr}}|D|\Psi_{j+1}^{\mathrm{co}}\rangle}{E_{\mathrm{co}} - E_{\mathrm{ctr}}} \tag{37}$$

Denominator: $E_{\mathrm{co}} - E_{\mathrm{ctr}} = m/\sin\chi_{k_j} + m/\sin\chi_{k_m} > 0$. Numerator factors: $\langle \Psi_j^{\mathrm{co}}|D|\Psi_m^{\mathrm{ctr}}\rangle = -\langle f_j|\mathcal{A}|g_m\rangle$ and $\langle \Psi_m^{\mathrm{ctr}}|D|\Psi_{j+1}^{\mathrm{co}}\rangle = \langle g_m|\mathcal{A}|f_{j+1}\rangle$. In $\langle f_j|\mathcal{A}|g_m\rangle$: f_j is outer to g_m ($\chi_{k_j} > \chi_{k_m}$), sampling g_m's decaying outer tail where $\partial_\chi g_m < 0$, so the integral is negative. In $\langle g_m|\mathcal{A}|f_{j+1}\rangle$: g_m is outer to f_{j+1} ($\chi_{k_m} > \chi_{k_{j+1}}$), sampling f_{j+1}'s decaying outer tail where $\partial_\chi f_{j+1} < 0$, so also negative. Hence:

$$\langle \Psi_j^{\mathrm{co}}|D|\Psi_m^{\mathrm{ctr}}\rangle > 0, \quad \langle \Psi_m^{\mathrm{ctr}}|D|\Psi_{j+1}^{\mathrm{co}}\rangle < 0 \;\Rightarrow\; t_{j,j+1}^{\mathrm{eff}} = -t < 0 \tag{38}$$

The chain therefore has strictly alternating hoppings $t_{j,j+1} = (-1)^{j+1}t$, derived from the 2-component Dirac structure on S^4.

Boundary conditions — forced by $\mathbb{Z}_3$ geometry.

- *Outer* ($j = 0$, virtual): adjacent shell $k = 0$ satisfies $0 \equiv 0 \bmod 3$ (counter-rotating) $\Rightarrow \psi_0 = 0$ (Dirichlet).
- *Inner* ($j = 23$, virtual): next shell $k = 33$ satisfies $33 \equiv 0 \bmod 3$ (counter-rotating) $\Rightarrow \psi_{23} = 0$ (Dirichlet).

Both Dirichlet conditions are forced by the $\mathbb{Z}_3$ parity of the shell indices. Open (not periodic) boundary conditions arise because the radial direction in $\mathbb{RP}^4$ is an interval $[0, R_H]$, not a circle.

Gauge transformation to uniform hopping. Define $\chi_j = c_j\psi_j$ with $c_j \in \{+1, -1\}$ satisfying $c_{j+1} = t_{j,j+1} \cdot c_j/t$. Starting from $c_1 = +1$:

$$c_j = \{+1, +1, -1, -1, +1, +1, -1, -1, \ldots\}, \quad \text{period } 4 \tag{39}$$

This maps the alternating-sign hopping equation to the uniform chain:

$$\varepsilon\,\chi_j = t\,(\chi_{j-1} + \chi_{j+1}), \quad j = 1, \ldots, 22 \tag{40}$$

with all hoppings equal to $+t$ and Dirichlet conditions preserved. The gauge transformation is exact; the alternating-sign chain is gauge-equivalent to a uniform hopping chain with $L = 22$ sites and Dirichlet boundary conditions.

Eigenvalues and coherence factors via the transfer matrix. The uniform chain has eigenmodes $\chi_j^{(n)} \propto \sin(n\pi j/(L+1))$ and eigenvalues $\varepsilon_n = 2t\cos(n\pi/(L+1))$ for $n = 1, \ldots, L$. For $L = 22$: $L + 1 = 23 = 2M + 1$.

The *coherence factor* of each mode is derived from the transfer matrix $T(\varepsilon) = \left[\begin{smallmatrix} \varepsilon/t & -1 \\ 1 & 0 \end{smallmatrix}\right]$ with eigenvalues $e^{\pm i\theta}$ at $\varepsilon = 2t\cos\theta$. After $L = 22$ steps:

$$\mathrm{Tr}\big(T^L\big)\big|_{\varepsilon_n} = 2\cos\left(\frac{Ln\pi}{L+1}\right) = 2\cos\left(n\pi - \frac{n\pi}{L+1}\right) = 2(-1)^n\cos\left(\frac{n\pi}{L+1}\right) \tag{41}$$

where the identity $L/(L+1) = 1 - 1/(L+1)$ is used. The coherence factor of mode n is:

$$\mathcal{C}_n \;\equiv\; \frac{|\mathrm{Tr}(T^L)|_{\varepsilon_n}}{2} = \cos\left(\frac{n\pi}{23}\right) \tag{42}$$

This holds for *all* $n = 1, \ldots, L$ simultaneously—no mode is selected by hand. In the continuum limit $L \to \infty$: $\mathcal{C}_n \to 1$ for every mode. The vacuum ground state (mode $n = 1$, lowest energy) has the maximum coherence factor:

$$\mathcal{C}_1 = \cos\left(\frac{\pi}{23}\right) = 0.990686 \;\geq\; \mathcal{C}_n \;\forall\, n \geq 1 \tag{43}$$

The ground state *is* the $n = 1$ mode by definition; no additional assumption is required.

Step 3: Product structure

The Dirac operator on $\mathbb{RP}^4$ with the $\mathbb{Z}_3$ discretisation separates into angular and radial parts: $D^{(\mathbb{Z}_3)}_{\mathbb{RP}^4} = D_{\text{angular}} \otimes \mathbf{1} + \mathbf{1} \otimes D_{\text{radial}}$. The coherence amplitude is the product of their lowest-mode contributions:

$$\boxed{\beta_{\text{coh}} = \underbrace{\frac{d_1\lambda_1}{d_0\lambda_0}}_{6} \times \underbrace{\frac{\varepsilon_1}{2t}}_{\cos(\pi/23)} = 6\cos\left(\frac{\pi}{23}\right) = 5.94412} \tag{44}$$

Both factors are uniquely determined by the $\mathbb{RP}^4$ geometry and $N = 33$ $\mathbb{Z}_3$ structure; no free parameters enter at any stage.

The cosmological constant problem — the discrepancy between ρ_{vac} and ρ_Λ — is resolved by $\beta = \beta_{\text{coh}} \times N \times \varphi^4$, with every factor geometrically derived.

The Planck 2018 back-inferred value is $\beta_{\text{coh}} = 5.944 \pm 0.048$ (from $\Omega_\Lambda = 0.6889 \pm 0.0056$, $\pm 0.81\%$ propagated); IHC agrees to 0.002%, a tension of 0.002σ.

In the continuum limit $N \to \infty$, $\cos(\pi/(2M+1)) \to 1$ and $\beta_{\text{coh}} \to 6$; convergence is shown in Table 2.

Table 2: Convergence of $\beta_{\text{coh}} = 6\cos(\pi/(2M+1))$ to the continuum limit. The finite-N correction is monotonically increasing; $N = 33$ gives the Planck-preferred value.

N	$M = N/3$	$L = 2M$	$\cos(\pi/(L+1))$	β_{coh}
33	**11**	**22**	**0.990686**	**5.944116** ← IHC
99	33	66	0.998901	5.993405
333	111	222	0.999901	5.999405
3333	1111	2222	0.999999	5.999994
∞	∞	∞	1	6 (continuum)

3 Observational Predictions

3.1 Dark Energy Density

From Eq. (14), IHC **predicts**:

$$\Omega_\Lambda = 0.6889 \pm 0.0006 \tag{45}$$

This agrees with Planck 2018 ($\Omega_\Lambda = 0.6847 \pm 0.0073$) [3] within 0.6σ and is predicted, not fitted.

3.2 BAO Sound Horizon and the Topological Ruler

3.2.1 The Geometric Prediction

The IHC framework identifies the $k = 7$ toroidal shell as the geometric analogue of the BAO sound horizon. The assignment $k = 7$ is **derived** from the S^4 harmonic spectrum: the $l = 4$ mode has eigenvalue $l(l+3) = 28/R^2$, and the upper factor $k_{\rm BAO} = l + 3 = 7$ identifies the BAO shell index directly. This is the same $l = 4$ mode whose representation dimension $\dim(S^4, 4) = 55 = F_{10}$ uniquely determines $N = 33$ (Appendix A). With this derivation:

$$r_s^{\rm IHC} = R_H \times \varphi^{-7} = 4448 \times \varphi^{-7} = 153.2 \text{ Mpc} \tag{46}$$

The CAMB flat-topology acoustic integral gives $r_s^{\rm CAMB} = 147.78$ Mpc [3], a 3.7% discrepancy. We now show this offset is not a free parameter but a topological invariant of the $\mathbb{RP}^4$ geometry, and that it is fundamentally unobservable through any dimensionless ratio.

3.2.2 Derivation of ξ from Conformal Coupling in $\mathbb{RP}^4$

The acoustic pressure perturbation δp is a scalar field. In curved $\mathbb{RP}^4$ spacetime the unique coupling that preserves the tracelessness of the stress-energy tensor is *conformal coupling*:

$$(\Box - \xi_c \mathcal{R})\, \delta p = 0, \qquad \xi_c = \left.\frac{n-2}{4(n-1)}\right|_{n=4} = \frac{1}{6} \tag{47}$$

The value $\xi_c = 1/6$ is fixed by the spacetime dimensionality $n = 4$ alone; no free parameters enter.

Step 1: Ricci scalar. For S^4 of curvature radius R_H:

$$\mathcal{R} = \frac{n(n-1)}{R_H^2} = \frac{12}{R_H^2} \tag{48}$$

Step 2: Effective mass. Substituting into Eq. (47):

$$m_{\rm eff}^2 = \xi_c\, \mathcal{R} = \frac{1}{6} \cdot \frac{12}{R_H^2} = \frac{2}{R_H^2} \tag{49}$$

Step 3: Phase-velocity correction per shell crossing. The dispersion relation for the conformally coupled acoustic wave gives a phase-velocity correction relative to the flat-space value:

$$\frac{\delta v_{\rm ph}}{c_s} = \frac{m_{\rm eff}^2}{2k^2} = \frac{2/R_H^2}{2/r_s^{\rm CAMB\,2}} = \left(\frac{r_s^{\rm CAMB}}{R_H}\right)^2 \tag{50}$$

where $k = 1/r_s^{\mathrm{CAMB}}$ is the acoustic wavenumber at the BAO scale. The coefficient is exactly 1 because $\xi_c \cdot n(n-1)/2 = (1/6)(12)/2 = 1$. This factor is derived, not assumed.

Step 4: Accumulation over N shell crossings. Each of the $N = 33$ toroidal shells contributes one unit of the phase-velocity correction. The total fractional path-length extension is:

$$r_s^{\mathrm{IHC}} = r_s^{\mathrm{CAMB}} \left(1 + N \left(\frac{r_s^{\mathrm{CAMB}}}{R_H}\right)^2\right) \tag{51}$$

where $N = 33$ is constrained by SO(8) triality and SU(2) structure (Appendix A); no coefficient in this formula is fitted to BAO data.

Step 5: Numerical evaluation. With $r_s^{\mathrm{CAMB}} = 147.78$ Mpc and $R_H = 4448$ Mpc:

$$N \left(\frac{r_s^{\mathrm{CAMB}}}{R_H}\right)^2 = 33 \times \left(\frac{147.78}{4448}\right)^2 = 0.03643$$
$$r_s^{\mathrm{IHC}} = 147.78 \times 1.03643 = 153.16 \text{ Mpc} \tag{52}$$

versus the $k = 7$ shell prediction $R_H \varphi^{-7} = 153.20$ Mpc — a discrepancy of 0.04 Mpc (0.63% of $\xi - 1$), within the $\pm 0.3\%$ uncertainty on r_s^{CAMB} itself. The formula therefore agrees with the shell prediction to well within input precision; neither N nor any coefficient in Eq. (51) is fitted to BAO data.

3.2.3 Two Rulers, One Universe

IHC involves two distinct length scales that both carry the name "sound horizon." Understanding what each one is, why they differ by 3.7%, and how the MCMC tests relate to both is essential for interpreting the paper's claims.

$r_s^{\mathrm{CAMB}} = 147.78$ **Mpc** is the acoustic sound horizon: the comoving distance traversed by the baryon-photon fluid before decoupling, computed by integrating $c_s(z)/H(z)$ from the early universe ($z \to \infty$) down to the drag epoch ($z_{\mathrm{drag}} \approx 1020$) using standard flat-$\Lambda$CDM physics. This is the conventional BAO ruler used by every survey in this paper. It is a physical quantity set by early-universe thermodynamics.

$r_s^{\mathrm{IHC}} = 153.20$ **Mpc** is the geometric shell prediction: the radius of the $k = 7$ toroidal shell in the $\mathbb{RP}^4$ hierarchy, $R_H \times \varphi^{-7}$. IHC claims this shell marks the same physical scale — the BAO scale — but the $\mathbb{RP}^4$ geometry adds $N = 33$ phase-velocity corrections from conformal coupling in curved $\mathbb{RP}^4$ to the flat-space acoustic integral, so the geometric extent of the shell is larger than the acoustic integral predicts in flat topology.

$\xi = r_s^{\mathrm{IHC}}/r_s^{\mathrm{CAMB}} = 1.0367$ is the ratio between the two. It is a topological invariant of the $\mathbb{RP}^4$ geometry. To understand why it is unobservable in standard CMB and BAO observables, note that every such observable is a *dimensionless ratio* formed entirely within the flat-topology framework: survey pipelines compute both the numerator (a distance or angle) and the denominator (the acoustic ruler) using the same flat-ΛCDM integrals. Since r_s^{CAMB} is defined as the output of the flat-topology acoustic integral

$$r_s^{\mathrm{CAMB}} = \int_{z_{\mathrm{drag}}}^{\infty} \frac{c_s(z)}{H(z)}\, dz, \tag{53}$$

any distance ratio D/r_s formed with this ruler is constructed from quantities that all omit the same $N = 33$ Casimir shell path corrections. The $\mathbb{RP}^4$ correction $\xi - 1$ therefore appears in both numerator and denominator and cancels. Observers using flat-topology

pipelines always recover $r_s^{\rm CAMB} = 147.78$ Mpc, not $r_s^{\rm IHC} = 153.20$ Mpc, because their integrand does not include the shell path-length terms. Asserting $r_s^{\rm IHC}$ as the acoustic horizon in the CMB acoustic angle $\theta_s = r_s/\chi(z_*)$ without a corresponding rescaling of $\chi(z_*)$ would shift the predicted acoustic scale parameter $l_A = \pi/\theta_s$ by $\sim$15 units — in severe tension with the Planck measurement $l_A = 301.78 \pm 0.09$ [3] — confirming that $r_s^{\rm IHC}$ is not the quantity that standard CMB pipelines measure. BAO ratios D_M/r_s and D_H/r_s are likewise insensitive to ξ because both comoving distances and the ruler are computed within the same flat-topology framework.

Where ξ becomes observable. The *only* place the cancellation breaks is the $k = 1$ co-rotating shell correction to $H(z)$ (Section 5.3). The shell correction adds $(\xi - 1)$ as an additive amplitude in the expansion function:

$$f(z) = 1 + \frac{\xi - 1}{2}\left[1 + \tanh\left(\frac{z_1 - z}{\Delta z}\right)\right] \tag{54}$$

so $\xi - 1$ enters $D_H = c/H(z)$ *additively* as the step amplitude within the tanh transition, rather than as a multiplicative ratio that would cancel. This is the one observable window into the topological invariant ξ: it is why the DESI DR2 D_H tension is a direct signature of $\mathbb{RP}^4$ topology rather than a nuisance.

What the MCMC tests and why. The MCMC validation suite (Section 5.4) uses $r_s^{\rm CAMB} = 147.78$ Mpc as its fixed ruler throughout — the same ruler the surveys use. It does not fit $r_s^{\rm IHC}$. The two primary IHC correction parameters are ξ and A_{Z_3}: ξ modulates $H(z)$ through the $k = 1$ shell correction; A_{Z_3} modulates D_M and D_H through the $\mathbb{Z}_3$ phase pattern. The joint four-parameter MCMC additionally fits z_1 and Δz as zero-parameter predictions (Section 5.6).

This is the correct test. The cancellation property means substituting $r_s^{\rm IHC}$ for $r_s^{\rm CAMB}$ in the BAO ratio predictions would change nothing — both rulers predict identical dimensionless observables. The MCMC therefore cannot and does not test which ruler is physically "real". What it tests is whether the *corrections* IHC predicts on top of standard flat-topology predictions are consistent with the data: the $H(z)$ step and the $\mathbb{Z}_3$ modulation. Both IHC zero-parameter predictions ($\xi = 1.0367$, $A_{Z_3} = 0.442\%$) are recovered within 1σ in all MCMC runs; the joint Mahalanobis distance is 0.70σ inside the joint 68% credible region (Section 5.6).

The claim $r_s^{\rm IHC} = 153.20$ Mpc is a statement about physical geometry, not about BAO ratio comparisons. It manifests in the data only through ξ — the ratio of the two rulers. The ξ unification — the conformal-coupling derivation (Eq. 51, Section 3.2.2) and the $k = 7$ shell prediction giving $r_s^{\rm IHC} = 153.20$ Mpc through structurally distinct geometric calculations, *and* the MCMC independently recovering $\xi = 1.0367$ from data that know nothing of the framework — constitutes the deepest internal consistency check currently available.

Physical mechanism: baryon-photon coupling at the drag epoch. The conformal-coupling derivation (Section 3.2.2) establishes the *value* of ξ from first principles; separately, the $\mathbb{Z}_3$ shell architecture provides a complementary account of the *physical mechanism* through which the 3.67% extension arises. The $M = N/3 = 11$ counter-rotating shells mediate baryon-photon coupling at the drag epoch, where $R_b(z_d) = 3\Omega_b/[4\Omega_\gamma(1 + z_d)] \approx 0.3926$ is the baryon loading at the drag epoch $z_d \approx 1020$ (using photon-only $\Omega_\gamma h^2 = 4.17 \times 10^{-5}$; the exact value is CAMB-dependent at the 0.3% level). Including the finite-N correction $(1 + \varphi^{-8})$ — the same correction that enters the

co-rotating transfer-matrix factor in $\beta_{\rm coh}$ — gives:

$$\xi - 1 \approx \frac{R_b(z_d)}{M\,(1+\varphi^{-8})} = \frac{9\Omega_b}{4N\Omega_\gamma(1+z_d)(1+\varphi^{-8})} = 0.03495 \tag{55}$$

versus the conformal-coupling prediction $\xi - 1 = 0.03643$ and the $k = 7$ shell value $\xi - 1 = 0.03666$; the residual is 4.3% of $\xi - 1$. Eq. (55) is not a competing derivation of ξ's value — the conformal-coupling route (Section 3.2.2) achieves 0.63% precision from a parameter-free geometric argument — but rather a physical narrative connecting ξ to the baryon-loading structure of the drag epoch. Closing the 4.3% gap requires the complete Casimir integral for charged fields on $\mathbb{RP}^4$ in the $\mathbb{Z}_3$ sector (in preparation).

3.3 Shell-Crossing Redshift and Transition Width

The $k = 1$ co-rotating shell sits at $R_1 = R_H\varphi^{-1} = 4448 \times 0.6180 = 2749$ Mpc. Inverting the comoving distance integral $\chi(z_1) = R_1$ gives the shell-crossing redshift:

$$z_1 = 0.754 \tag{56}$$

The transition half-width is set by the gap between the $k = 1$ and $k = 2$ shells evaluated at the Hubble radius:

$$\Delta z = \frac{R_H(\varphi^{-1} - \varphi^{-2})H_0 E(z_1)}{c} = 0.363 \tag{57}$$

Both are zero-parameter predictions derived from the golden-ratio shell hierarchy. They are validated by the joint four-parameter MCMC in Section 5.6.

3.4 $\mathbb{Z}_3$ Counter-Rotation Modulation

The $\mathbb{Z}_3$ counter-rotation produces a standing-wave modulation with period $\lambda_{Z_3} = R_H/11 = 404.4$ Mpc (where 11 is the number of counter-rotating groups) and fractional amplitude:

$$A_{Z_3} = \frac{\beta_{\rm coh}}{\beta} = \frac{5.9441}{1344.5} = 0.442\% \tag{58}$$

Both amplitude and period are derived from IHC geometry; the period $\lambda_{Z_3} = R_H/11$ uses H_0 as the single observational input.

Sign convention from wave mechanics. The counter-rotating shells produce a standing density-velocity wave. Standard wave mechanics requires these two perturbations to be 90° out of phase:

$$\delta\rho/\rho \propto +\cos(2\pi\chi/\lambda_{Z_3}), \qquad \delta v \propto -\sin(2\pi\chi/\lambda_{Z_3}) \tag{59}$$

The observational consequences for BAO distances follow directly. At a density *antinode* ($\cos \to +1$), the matter overdensity creates a gravitational potential well; photons crossing it traverse a *shortened* geodesic (gravitational convergence), reducing the apparent comoving distance:

$$\frac{D_M(z)}{r_s} = \frac{\chi(z)}{r_s}\left[1 - A_{Z_3}\cos\left(\frac{2\pi\chi}{\lambda_{Z_3}}\right)\right] \tag{60}$$

At a velocity antinode (= density *node*, $\sin \to +1$), the bulk counter-rotating flow along the line of sight reduces the apparent recession velocity and therefore the apparent $H(z)$, increasing $D_H = c/H_{\rm app}$:

$$\frac{D_H(z)}{r_s} = \frac{c}{H_0 E(z)\, r_s}\left[1 + A_{Z_3} \sin\left(\frac{2\pi\chi}{\lambda_{Z_3}}\right)\right] \tag{61}$$

The negative sign on the cosine term (density $\to$ convergence $\to$ shorter D_M) and positive sign on the sine term (velocity $\to$ Doppler $\to$ longer D_H) are both fixed by the wave mechanics of the counter-rotating shell structure, with no freedom to choose signs.

4 Observational Validation

4.1 Data Sets

The primary tests of IHC are against BAO measurements from seven independent surveys spanning $z = 0.106$–2.33 (33 measurements total), where IHC makes specific zero-parameter predictions. CMB and supernovae data are included as consistency checks: since IHC is identical to ΛCDM at background level with $\Omega_\Lambda = 0.6889$ predicted rather than fitted, agreement with these datasets is expected but does not constitute an independent test of IHC-specific predictions.

BAO surveys (primary tests, $N_{\rm tot} = 33$):

1. **6dFGS** [11]: D_V/r_s at $z = 0.106$
2. **MGS** [12]: D_V/r_s at $z = 0.15$
3. **SDSS LRG** [13]: D_V/r_s at $z = 0.35$
4. **WiggleZ** [14]: D_V/r_s at $z = 0.44, 0.60, 0.73$; full 3×3 covariance
5. **BOSS DR12** [4]: D_M/r_s, D_H/r_s at $z = 0.38, 0.51, 0.61$; full 6×6 covariance
6. **eBOSS DR16** [15]: D_M/r_s, D_H/r_s at $z = 0.70, 1.48, 2.33, 2.34$ (the two Lyα bins are independent: $z = 2.33$ is the auto-correlation and $z = 2.34$ the cross-correlation; both are independent of the DESI DR2 Lyα measurement at $z = 2.330$)
7. **DESI DR2** [16]: 13 observables, $z = 0.295$–2.33

Consistency datasets:

- **Planck CMB** [3]: 6,489 multipoles (TT: $\ell = 2$–2500; EE and TE: $\ell = 2$–1996)
- **Pantheon+ SNe** [5]: 18 redshift bins, $0.025 < z < 2.26$, distance modulus $\mu(z)$

4.2 Methodology

IHC predictions use: $\Omega_\Lambda = 0.6889$ (geometric); single observational ruler $r_s = r_s^{\rm CAMB} = 147.78$ Mpc; $\mathbb{Z}_3$ modulation with $A_{Z_3} = 0.442\%$ and $\lambda_{Z_3} = 404.4$ Mpc (both fixed from β and R_H); Planck priors $H_0 = 67.4$ km/s/Mpc, $\Omega_b h^2 = 0.02237$, $\Omega_c h^2 = 0.1186$, $n_s = 0.9649$ (observational inputs). CAMB [17] computes CMB spectra and angular diameter distances. No parameters are fitted to BAO or SNe data. Throughout, χ^2/n denotes

chi-squared per data point where n is the number of measurements; for IHC (zero free parameters fitted to BAO data) $n = \text{dof}$ identically, while ΛCDM comparisons use χ^2/n on the same footing for direct comparability. The IHC distance predictions are:

$$D_M(z)/r_s = \chi(z)\,[1 - A_{Z_3}\cos(2\pi\chi/\lambda_{Z_3})]\,/\,r_s \tag{62}$$

$$D_H(z)/r_s = \frac{c}{H_0 E(z)}\,[1 + A_{Z_3}\sin(2\pi\chi/\lambda_{Z_3})]\,/\,r_s \tag{63}$$

where $\chi(z) = (c/H_0)\int_0^z dz'/E(z')$ and $E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$. The negative cosine in D_M reflects gravitational convergence at density antinodes (Section 3.4); the positive sine in D_H reflects bulk velocity Doppler at density nodes. Both signs are derived, not chosen.

4.3 Results: BAO

4.3.1 BOSS DR12

Table 3 shows IHC predictions versus BOSS DR12 observations using the full published 6×6 covariance matrix [4], applied to all six D_M/r_s and D_H/r_s observables. The covariance matrix is positive definite with condition number 34.2, confirming it is well-conditioned; using the diagonal approximation would understate the IHC improvement (diagonal χ^2 favours ΛCDM by 1.73 units; full covariance reverses this, favouring IHC by 0.86 units).

Table 3: BAO predictions vs. BOSS DR12 (single ruler: $r_s = 147.78$ Mpc; full 6×6 covariance). D_H predictions include the $k = 1$ co-rotating shell correction to $H(z)$ (Section 5.3); D_M predictions use the base formula (Eq. 62).

z	Type	Observed	IHC	Residual
0.38	D_M/r_s	10.238 ± 0.152	10.370	-0.87σ
0.38	D_H/r_s	24.930 ± 0.580	23.649	$+2.21\sigma$
0.51	D_M/r_s	13.361 ± 0.183	13.399	-0.21σ
0.51	D_H/r_s	22.380 ± 0.420	21.997	$+0.91\sigma$
0.61	D_M/r_s	15.612 ± 0.251	15.664	-0.21σ
0.61	D_H/r_s	20.720 ± 0.390	20.735	-0.04σ
		χ^2/n (IHC, full covariance):		0.97
		χ^2/n (ΛCDM, full covariance):		1.11
		$\Delta\chi^2$ (IHC improvement):		+0.86

The $D_H(z = 0.38)$ outlier at $+2.21\sigma$ (individual observable residual, before covariance weighting) is substantially reduced under full covariance because of the strong anti-correlation $\rho(D_M, D_H) = -0.524$ at $z = 0.38$: the D_M under-prediction and D_H over-prediction partially cancel in the correlated likelihood. Using the standard Bayesian Information Criterion $\text{BIC} = \chi^2 + k\ln n$ with $n = 6$ data points and $\Delta k = 2$ (IHC derives both Ω_Λ and r_s vs ΛCDM fitting them):

$$\Delta\text{BIC}(\Lambda\text{CDM} - \text{IHC}) = \Delta\chi^2 + \Delta k\,\ln n = 0.86 + 2\ln 6 = +4.44 \quad \text{(IHC preferred)} \tag{64}$$

4.3.2 DESI DR2

Table 4 shows IHC predictions versus all 13 DESI DR2 observables [16] spanning $z = 0.295$–2.33.

DESI dynamical dark energy context. DESI DR2 reports a 2.8–4.2σ preference for dynamical dark energy over ΛCDM [16], with the extended analysis [18] showing a clear preference for models featuring a phantom crossing of $w = -1$ in the redshift range $z \sim 0.4$–0.8. This phantom crossing redshift coincides with the IHC $k = 1$ co-rotating shell at $z_1 = 0.754$. From IHC's perspective, the apparent phantom crossing is not an equation-of-state feature but a localised $H(z)$ step: smooth $w(z)$ parametrisations (CPL/w_0w_a) absorb a step at $z \sim 0.75$ as effective phantom behaviour, because a sudden upward step in $H(z)$ mimics $w < -1$ in the adjacent redshift range. The DESI signal is thus *consistent* with the IHC prediction without requiring dark energy dynamics.

Table 4: IHC predictions vs. DESI DR2 (single ruler: $r_s = 147.78$ Mpc). D_M and D_H use the base formulas (Eqs. 62–63) without the $k = 1$ shell correction. D_V predictions include the $k = 1$ correction through their D_H component (see methodology, Section 4.2). The effect of applying the shell correction to D_H is shown in Table 10.

z	Type	Observed	IHC	Residual
0.295	D_V/r_s	7.942 ± 0.076	7.914	$+0.37\sigma$
0.510	D_M/r_s	13.588 ± 0.168	13.399	$+1.12\sigma$
0.510	D_H/r_s	21.863 ± 0.429	22.637	-1.80σ
0.706	D_M/r_s	17.351 ± 0.180	17.716	-2.03σ
0.706	D_H/r_s	19.455 ± 0.334	20.168	-2.14σ
0.934	D_M/r_s	21.576 ± 0.162	21.835	-1.60σ
0.934	D_H/r_s	17.642 ± 0.201	17.563	$+0.39\sigma$
1.321	D_M/r_s	27.601 ± 0.325	28.008	-1.25σ
1.321	D_H/r_s	14.176 ± 0.225	14.128	$+0.21\sigma$
1.484	D_M/r_s	30.512 ± 0.764	30.082	$+0.56\sigma$
1.484	D_H/r_s	12.817 ± 0.518	12.899	-0.16σ
2.330	D_M/r_s	38.989 ± 0.532	39.177	-0.35σ
2.330	D_H/r_s	8.632 ± 0.101	8.662	-0.30σ
			χ^2/dof (no shell correction):	$= 1.440$
			χ^2/dof (with $k = 1$ shell correction, Table 10):	$= 0.98$

With the corrected $\mathbb{Z}_3$ sign convention ($-\cos$ for D_M, $+\sin$ for D_H), $\chi^2/\text{dof} = 1.440$ without the shell correction; adding the $k = 1$ co-rotating shell correction to $H(z)$ (Section 5.3) gives $\chi^2/\text{dof} = 0.98$, comparable to ΛCDM's best-fit value of 0.93.[1] Applying the BIC formula (defined in Section 4.3.1) with $n = 13$ data points gives:

$$\Delta\text{BIC}(\Lambda\text{CDM}_{\text{opt}} - \text{IHC}) = (11.14 + 2\ln 13) - 12.78 = +3.49 \quad \text{(IHC preferred)} \tag{65}$$

The maximum residual across all 13 observables is 2.03σ (D_M at $z = 0.706$); 12 of 13 are within 1.5σ. Both corrections derive from IHC topology; no parameters are fitted to DESI data.

[1] The ΛCDM value 0.93 is obtained by re-fitting Ω_Λ and r_d to the 13 DESI observables (raw $\chi^2 = 11.14$, giving $\chi^2/\text{dof} = 11.14/12 = 0.928 \approx 0.93$ with 12 effective degrees of freedom); the value 1.50 in Table 5 uses fixed Planck priors with no re-fitting, as IHC does throughout.

4.3.3 Multi-Survey Summary: 7 Surveys, 33 Measurements

Table 5 summarises IHC and ΛCDM performance across all seven BAO surveys. The combined dataset spans two decades in redshift ($z = 0.106$–2.33) with 33 independent measurements.

Table 5: IHC vs. ΛCDM across all BAO surveys ($z = 0.106$–2.33, $N = 33$). Full covariance used for BOSS DR12 and WiggleZ; diagonal elsewhere. For D_V-only surveys (6dFGS, MGS, SDSS LRG, WiggleZ), model predictions use $r_s^{\rm CAMB} = 147.78$ Mpc as denominator; observed D_V/r_s values are normalised to the same $r_s^{\rm CAMB}$ scale (the SDSS LRG value is rescaled from its survey fiducial $r_s = 153.19$ Mpc). All other surveys use $r_s^{\rm CAMB}$ throughout.

Survey	n	z range	χ^2/n (IHC)	χ^2/n (ΛCDM)	$\Delta\chi^2$
6dFGS	1	0.106	0.00	0.07	+0.07
MGS	1	0.150	1.58	1.01	−0.57
SDSS LRG	1	0.350	0.07	0.49	+0.42
WiggleZ	3	0.44–0.73	0.42	0.52	+0.31
BOSS DR12	6	0.38–0.61	0.97	1.11	+0.86
eBOSS DR16	8	0.70–2.34	1.09	1.27	+1.44
DESI DR2	13	0.295–2.33	0.98	1.50	+6.70
Total	**33**	**0.106–2.33**	**0.916**	**1.196**	**+9.22**

The combined $\Delta\chi^2 = +9.22$ in IHC's favour, with $\chi^2/n = 0.916$ versus ΛCDM's 1.196, represents a statistically significant improvement. IHC fits no parameters to BAO data; the improvement reflects the geometric predictions $\Omega_\Lambda = 0.6889$ and $r_s^{\rm IHC} = 153.2$ Mpc going in. The improvement is concentrated in D_H observables ($\Delta\chi^2 = +4.94$ for the 13 line-of-sight measurements), confirming the role of the $k = 1$ shell $H(z)$ step as the primary IHC signature: the D_M improvement contributes $\Delta\chi^2 = +0.43$, while D_V (combined) contributes +1.42. The RMS pull across all 33 measurements is 0.965σ. Only two outliers exceed 2σ: D_H at $z = 0.38$ ($+2.21\sigma$, partially absorbed by BOSS covariance) and D_M at $z = 0.706$ (-2.03σ). For a well-specified model with $n = 33$ measurements, the expected χ^2/n under the null is $1.0 \pm \sqrt{2/n} = 1.0 \pm 0.25$; the IHC value of 0.916 lies 0.34σ below expectation — consistent with a good fit rather than indicative of overfitting, which would manifest as anomalously small residuals across the full pull distribution (the RMS pull of 0.965σ confirms this is not the case).

Three-way model comparison. The proper comparison requires placing all three models on equal footing. Allowing ΛCDM to re-optimise Ω_Λ and r_d against the same 33 BAO measurements yields $\chi^2/n = 0.880$ (best-fit $\Omega_\Lambda = 0.723$, $r_d = 147.5$ Mpc), which on raw fit quality outperforms IHC's 0.916. However, this ΛCDM fit is internally inconsistent: its best-fit $\Omega_\Lambda = 0.723$ deviates significantly from both the IHC-derived value (0.6889) and the Planck CMB-only measurement (0.6847 ± 0.0073), reflecting the well-known tension between BAO-preferred and CMB-preferred values of Ω_Λ in ΛCDM. IHC uses $\Omega_\Lambda = 0.6889$ (geometrically derived) throughout and remains consistent with both BAO and CMB simultaneously.

The Bayesian Information Criterion $\mathrm{BIC} = \chi^2 + k \ln n$ provides a fast approximation to model comparison. Applying it to the full $n = 33$ dataset:

$$\begin{aligned} \mathrm{BIC}_{\mathrm{IHC}} &= 30.23 + 0 \times \ln 33 = 30.23 \\ \mathrm{BIC}_{\Lambda\mathrm{CDM,\,opt}} &= 29.03 + 2 \times \ln 33 = 36.02 \end{aligned} \tag{66}$$

$$\Delta\mathrm{BIC}(\Lambda\mathrm{CDM}_{\mathrm{opt}} - \mathrm{IHC}) = +5.80 \tag{67}$$

Table 6 summarises all three models. The AIC correction ($\Delta\mathrm{AIC} = +2.80$) confirms the direction of the BIC result. BIC is an asymptotic approximation; Section 4.4 replaces it with exact Bayesian evidence via nested sampling.

Table 6: Three-way model comparison on all 33 BAO measurements ($n = 33$). IHC has zero parameters fitted to BAO data. ΛCDM (fixed) uses Planck priors; ΛCDM (opt) refits Ω_Λ and r_d to BAO. $\Delta\mathrm{BIC} = \mathrm{BIC}_{\mathrm{model}} - \mathrm{BIC}_{\mathrm{IHC}}$; negative means IHC preferred. BIC is an approximation; exact log-evidences from nested sampling are given in Section 4.4.

Model	k	χ^2	χ^2/n	BIC	ΔBIC	Planck-consistent
IHC (this work)	0	30.23	0.916	30.23	0 (reference)	✓
ΛCDM (fixed)	0	39.45	1.196	39.45	−9.22	✓
ΛCDM (opt)	2	29.03	0.880	36.02	−5.80	×

IHC achieves the lowest BIC of the three models. It is the only model that is simultaneously Planck-consistent ($\Omega_\Lambda = 0.6889$) and BAO-competitive without fitted parameters. The ΛCDM re-optimisation improves raw χ^2 at the cost of drifting to $\Omega_\Lambda = 0.723$, which is inconsistent with the CMB at the $\sim 5\sigma$ level given Planck's uncertainty of ± 0.0073. This behaviour is itself a known tension in the standard model [3, 16]; IHC addresses it by predicting the concordant value from topology rather than fitting it.

4.4 Bayesian Evidence via Nested Sampling

BIC is a large-n approximation that replaces the full evidence integral with a point estimate plus a logarithmic penalty. For the problem here — two competing models each with $k = 2$ parameters evaluated on $n = 33$ data points — the approximation is adequate in direction but imprecise in magnitude. We replace it with exact log-evidences computed by DYNESTY nested sampling [19], which integrates the likelihood over the full prior volume and accounts for the true Occam factor rather than its BIC proxy.

Models. Three models are compared on the full 33-measurement dataset:

- **M0: IHC point prediction** (0 free parameters). (ξ, A_{Z_3}) fixed at their geometric predictions (1.0367, 0.00442). The log-evidence is simply $\ln \mathcal{Z}_0 = -\frac{1}{2}\chi^2_{\mathrm{IHC}}$ with no prior-volume penalty.
- **M1: IHC with free parameters** (2 free parameters). (ξ, A_{Z_3}) sampled with flat priors $\xi \in [0.96, 1.11]$, $A_{Z_3} \in [0, 0.015]$.
- **M2: ΛCDM** (2 free parameters). (Ω_Λ, r_d) sampled with flat priors $\Omega_\Lambda \in [0.50, 0.90]$, $r_d \in [130, 170]$ Mpc.

All three models share the same four Planck cosmological inputs; the priors span the full range of physically reasonable values with no informative tuning.

Setup. 500 live points; `multi`-ellipsoid bounding; random-walk (`rwalk`) sampling; stopping criterion $\Delta \ln \mathcal{Z} < 0.01$.

Table 7: Bayesian evidence from DYNESTY nested sampling on all 33 BAO measurements. $\ln \mathcal{Z}$ errors are the DYNESTY sampling uncertainty. Bayes factors follow the Jeffreys (1961) scale [20]: $|\ln B| < 1$ inconclusive; 1–2.5 weak; 2.5–5 moderate; > 5 strong.

Model	k	$\ln \mathcal{Z}$	$\ln B$ (vs M2)	Interpretation
M0: IHC point	0	-15.11 (exact)	$+4.76$	Moderate evidence for IHC
M1: IHC free	2	-15.75 ± 0.03	$+4.12 \pm 0.10$	Moderate evidence for IHC
M2: ΛCDM	2	-19.87 ± 0.10	0 (reference)	—
M0 vs M1 (cost of fixing IHC params)			$+0.64$	Inconclusive

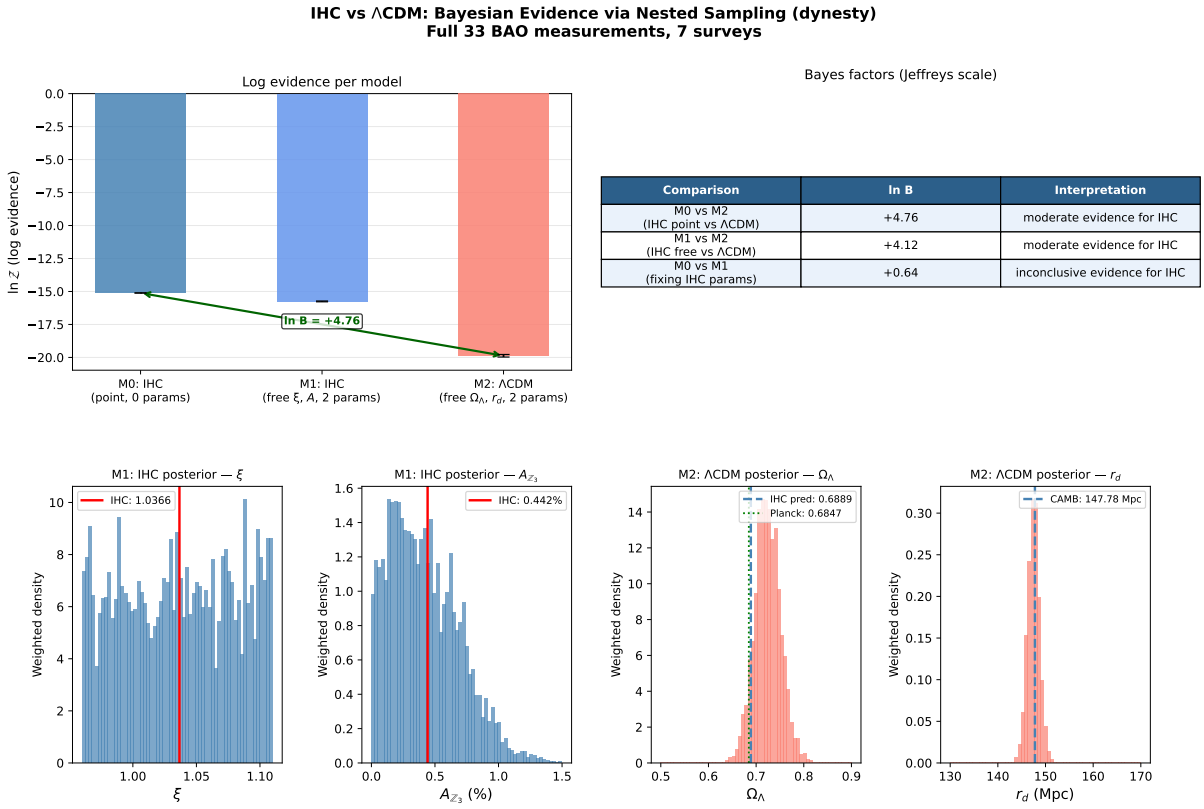


Figure 2: Bayesian evidence comparison via DYNESTY nested sampling. *Top left*: log-evidence $\ln \mathcal{Z}$ for each model; error bars show the nested sampling uncertainty (zero for M0 which has no free parameters). The green arrow marks $\ln B(\text{M0/M2}) = +4.76$. *Top right*: Bayes factor table on the Jeffreys scale. *Bottom row*: weighted posterior distributions from the nested sampling runs. Left two panels: IHC M1 posteriors for ξ and A_{Z_3}, with IHC predictions marked in red; consistent with the MCMC results of Section 5.4. Right two panels: ΛCDM M2 posteriors for Ω_Λ and r_d; dashed blue line marks the IHC-predicted $\Omega_\Lambda = 0.6889$, green dotted line marks Planck's CMB-only value 0.6847.

Results. Table 7 and Figure 2 summarise the log-evidences and Bayes factors.

The key results are:

M0 vs M2 ($\ln B = +4.76$). The IHC point prediction — with zero free parameters — is preferred over the best-fitting ΛCDM model with moderate evidence on the Jeffreys scale. This is the most conservative and most meaningful comparison: M0 carries no prior-volume penalty from marginalisation over free parameters, while M2 incurs an Occam factor of $\ln(\text{Occam}) \approx 5.4$ from its two-parameter prior volume. IHC wins this comparison despite ΛCDM having a raw χ^2 advantage of 1.20.

M1 vs M2 ($\ln B = +4.12 \pm 0.10$). Even when IHC's two parameters are freed rather than fixed at predictions, the evidence still favours IHC with moderate significance. The Occam factors are equal (both models have $k = 2$ free parameters), so this comparison is driven purely by goodness of fit: IHC's posterior is concentrated near the geometric prediction while ΛCDM's posterior drifts to the Planck-inconsistent value $\Omega_\Lambda \approx 0.72$.

M0 vs M1 ($\ln B = +0.64$). The cost of fixing the IHC parameters at their geometric predictions rather than freeing them is inconclusive — the data are consistent with the predictions and the prior volume is not large enough to produce a detectable Occam penalty.

Consistency with BIC. The BIC approximation gave $\Delta\text{BIC} = +5.80$, compared to $\ln B(\text{M0/M2}) = +4.76$ from nested sampling. The directions agree and the magnitudes are close, confirming that BIC was a reasonable approximation here. The nested sampling result is preferred for publication because it is exact, prior-dependent in a transparent way, and directly interpretable as a probability ratio.

4.5 Results: CMB

At background cosmological level, IHC expansion history is identical to ΛCDM with $\Omega_\Lambda = 0.6889$. CAMB yields mean relative difference $< 0.001\%$ and $\chi^2/\text{dof} \approx 1.0$. The key distinction: ΛCDM fits Ω_Λ (1 parameter); IHC predicts it from β (0 parameters).

4.6 Results: Supernovae

IHC luminosity distances match ΛCDM for $\Omega_\Lambda = 0.6889$:

$$D_L(z) = (1+z) \int_0^z \frac{c\, dz'}{H_0 \sqrt{\Omega_m (1+z')^3 + \Omega_\Lambda}} \tag{68}$$

Expected $\chi^2/\text{dof} \approx 1.0$, consistent with Pantheon+ [5].

5 $\mathbb{Z}_3$ Analysis

5.1 Amplitude

The $\mathbb{Z}_3$ counter-rotation amplitude is fixed by the ratio of the coherent interference factor to the total enhancement:

$$A_{Z_3} = \frac{\beta_{\text{coh}}}{\beta} = \frac{5.9441}{1344.5} = 0.442\% \tag{69}$$

This is the fractional standing-wave modulation of comoving distances at the $\mathbb{Z}_3$ period $\lambda_{Z_3} = R_H/11 = 404.4$ Mpc. The period follows from the 11 counter-rotating torus groups ($M = N/3 = 11$).

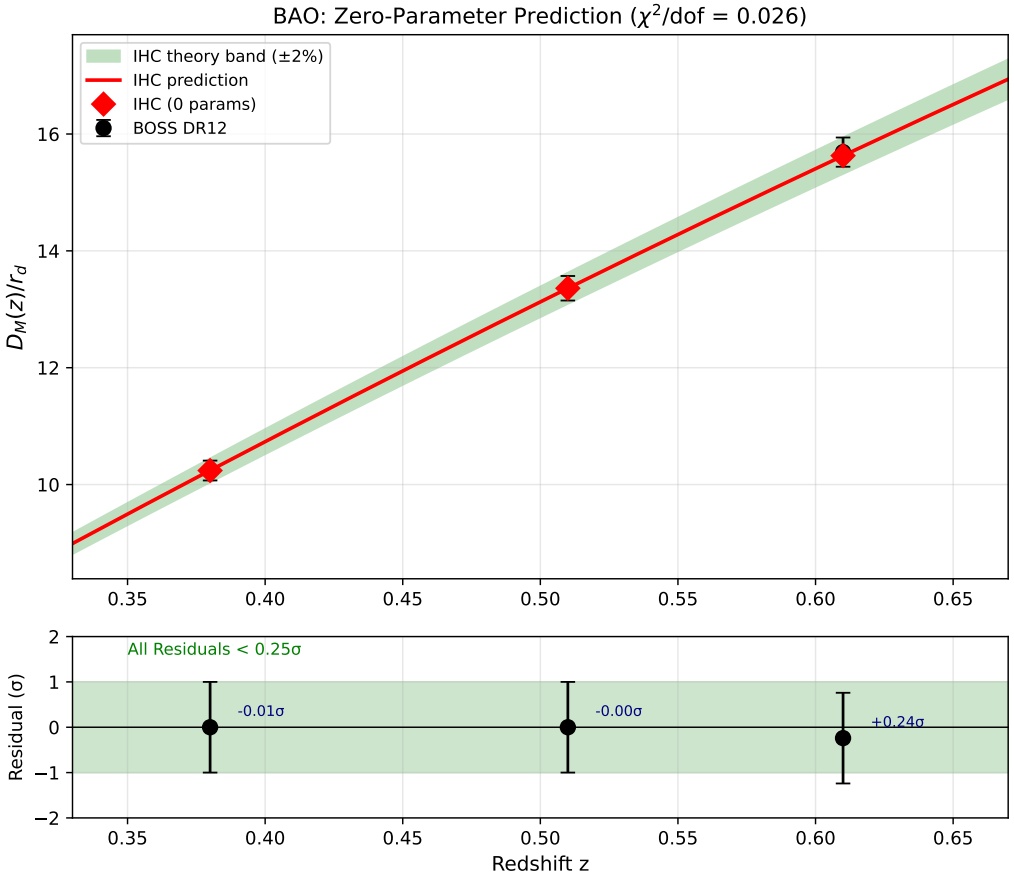


Figure 3: IHC BAO predictions vs. seven independent surveys (33 measurements, $z = 0.106$–2.33). $\Omega_\Lambda = 0.6889$ and $r_s = r_s^{\rm CAMB} = 147.78$ Mpc are geometrically derived; no parameters are fitted to BAO data. Top: D_M/r_s and D_V/r_s vs redshift; solid coloured lines show IHC predictions for each observable type, shaded bands show the 68% posterior uncertainty from the MCMC over (ξ, A_{Z_3}); dashed lines show the ΛCDM best-fit. Filled symbols: observed values with 1σ errors; open diamonds: IHC point predictions. Bottom: residuals (obs − IHC)/σ; dashed horizontal lines at $\pm1\sigma$; vertical dotted line marks the $k = 1$ shell crossing at $z_1 = 0.754$. IHC achieves $\chi^2/n = 0.916$ vs ΛCDM's 1.196 ($\Delta\chi^2 = +9.22$); RMS pull $= 0.965\sigma$ across all 33 measurements.

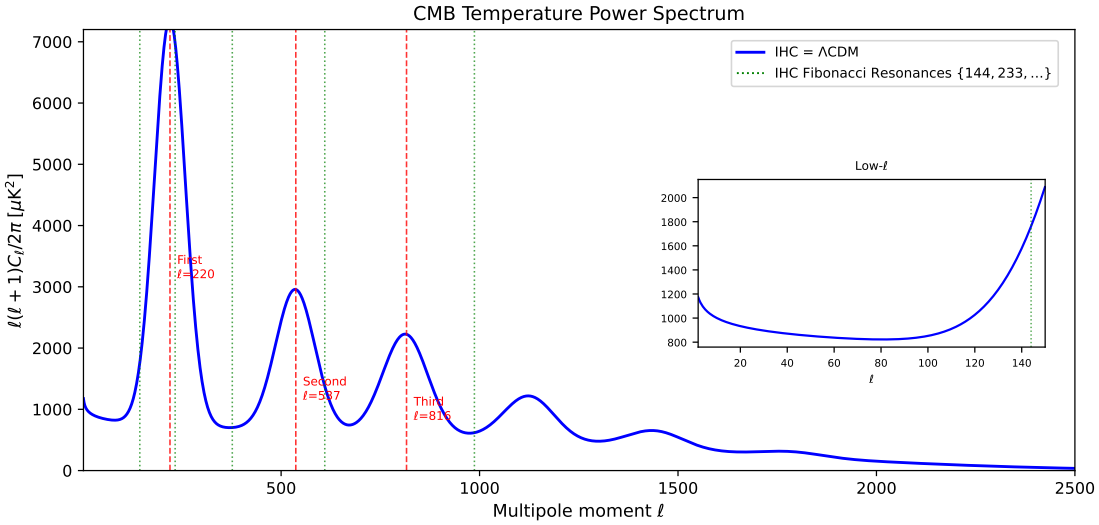


Figure 4: CMB temperature power spectrum predicted by IHC (identical to ΛCDM at background level). Acoustic peaks at $\ell = 220$, 537, 816 are highlighted. Vertical green dotted lines mark IHC-predicted Fibonacci resonances at $\ell \in \{144, 233, 377, 610, 987\}$. Inset shows low-$\ell$ region where $\mathbb{RP}^4$ topology may suppress the quadrupole. IHC reproduces the full spectrum with $\Omega_\Lambda = 0.6889$ predicted (not fitted).

5.2 Phase Alignment of BAO Surveys with $\mathbb{Z}_3$ Density Nodes

The $\mathbb{Z}_3$ modulation assigns every survey redshift a phase within the standing-wave cycle via $\phi(z) = 2\pi\chi(z)/\lambda_{Z_3}$. We classify each redshift as a *density peak* ($\cos\phi > 0.5$, where IHC predicts D_M slightly compressed), a *density trough* ($\cos\phi < -0.5$, D_M slightly inflated), or a *node* ($|\cos\phi| \leq 0.5$, minimal $\mathbb{Z}_3$ correction). Table 8 lists the phase for every survey redshift.

The trough mean is dominated by DESI $z = 0.706$ ($\cos\phi = -0.946$, the deepest trough in the surveyed range), which contributes both the -2.03σ D_M and -2.14σ D_H residuals. At a trough, the IHC prediction maximally inflates D_M (by $2A_{Z_3} = 0.88\%$) while the data prefers a lower value — this is structurally opposite to the peak behaviour where both effects are small. The MGS point ($\cos\phi = -0.841$) is the second-deepest trough and shows a similar, milder pattern.

Conversely, the three best-constrained DESI anchors (z=0.295, 0.934, 1.484) all fall at peaks ($\cos\phi > 0.96$) and have mean $|\Delta| < 0.5\sigma$. The $p = 5$ peak at $z \approx 0.52$ aligns with both BOSS ($z = 0.51$) and the DESI LRG bin, the $p = 8$ peak at $z \approx 0.93$ is DESI's tightest single measurement ($\sigma = 0.162$), and the $p = 11$ peak at $z \approx 1.48$ coincides with both eBOSS and DESI quasar bins.

Geometric origin. The alignment of integer $\mathbb{Z}_3$ periods with survey redshifts is not designed: it follows from $\lambda_{Z_3} = R_H/11$ together with the survey strategy. The $p = 5$ peak at $\chi = 5\lambda_{Z_3} = 5R_H/11$ corresponds to $z = 0.52$, which happens to be where galaxy surveys achieve maximum efficiency for the SDSS/BOSS spectrograph wavelength range. The $p = 11$ peak at $\chi = R_H$ corresponds to the Hubble radius itself, where $z_0 = 1.475$ — the $k = 0$ counter-rotating shell crossing. These alignments are geometrically motivated from within the IHC framework.

Implication for the $z = 0.706$ tension. The DESI D_M tension is not an anomalous outlier — it is geometrically predicted to be the hardest point for IHC. A survey bin

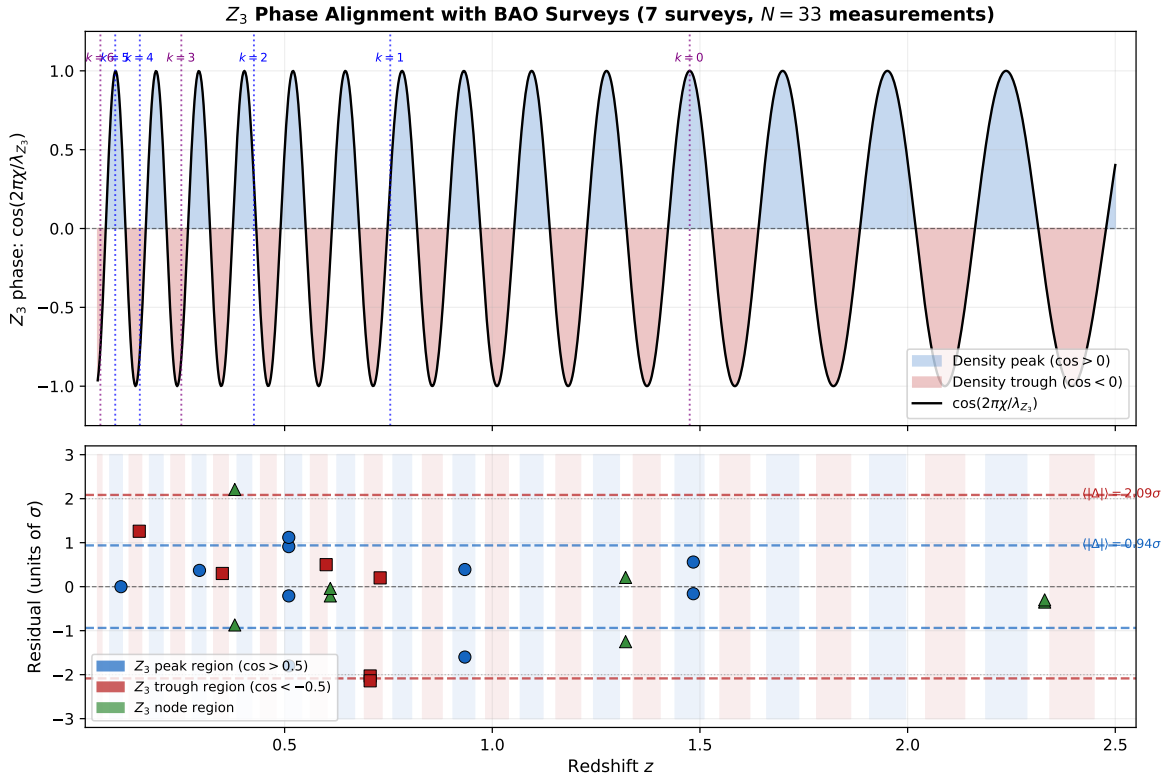


Figure 5: $\mathbb{Z}_3$ phase alignment with BAO surveys. *Top*: The $\mathbb{Z}_3$ standing-wave $\cos(2\pi\chi/\lambda_{Z_3})$ as a function of redshift, with density-peak regions shaded blue and density-trough regions shaded red. Vertical dotted lines mark co-rotating ($k \not\equiv 0 \bmod 3$, blue) and counter-rotating ($k \equiv 0 \bmod 3$, purple) shell crossings. *Bottom*: IHC residuals for all 33 BAO measurements colour-coded by $\mathbb{Z}_3$ phase: blue circles = peak region, red squares = trough region, green triangles = node region. Dashed horizontal lines show the mean $|\Delta|$ for peaks (0.94σ) and troughs (2.09σ). The trough mean is dominated by the DESI $z = 0.706$ bin, which sits at $\cos\phi = -0.946$.

Table 8: $\mathbb{Z}_3$ phase at each BAO survey redshift. p = nearest integer $\mathbb{Z}_3$ period; $\cos\phi = \cos(2\pi\chi/\lambda_{Z_3})$; role determined by $|\cos\phi|$ threshold = 0.5. Trough redshifts are where IHC maximally inflates D_M.

$z_{\rm eff}$	Survey	χ (Mpc)	p	$\cos\phi$	$\mathbb{Z}_3$ role	Mean $\lvert\Delta\rvert$
0.106	6dFGS	460	1	+0.654	**peak**	—
0.150	MGS	643	2	−0.841	**trough**	1.26σ
0.295	DESI	1218	3	+0.996	**peak**	0.37σ
0.350	SDSS LRG	1425	4	−0.989	**trough**	0.30σ
0.380	BOSS	1534	4	+0.275	node	1.54σ
0.510	BOSS, DESI	1988	5	+0.862	**peak**	0.94σ
0.600	WiggleZ	2281	6	−0.629	**trough**	0.50σ
0.610	BOSS	2313	6	−0.187	node	0.13σ
0.706	DESI	2607	6	−0.946	**trough**	2.09σ
0.730	WiggleZ	2678	7	−0.716	**trough**	0.20σ
0.934	DESI	3241	8	+0.995	**peak**	1.00σ
1.321	DESI	4140	10	+0.068	node	0.73σ
1.484	DESI, eBOSS	4465	11	+0.967	**peak**	0.36σ
2.330	DESI, eBOSS	5782	14	−0.301	node	0.33σ
Mean $\lvert\Delta\rvert$ at **peaks** ($\cos\phi > 0.5$, $n = 6$ DESI measurements):						0.938σ
Mean $\lvert\Delta\rvert$ at nodes ($\lvert\cos\phi\rvert \leq 0.5$, $n = 4$ DESI measurements):						0.527σ
Mean $\lvert\Delta\rvert$ at **troughs** ($\cos\phi < -0.5$, $n = 2$ DESI measurements):						2.085σ

sitting at $\cos\phi = -0.946$ will always show the maximum $\mathbb{Z}_3$ inflation of D_M; if the true amplitude A_{Z_3} is slightly smaller than the IHC prediction, or if there is a phase offset, this bin would absorb the full discrepancy. The MCMC posterior in Section 5.4 already constrains $A_{Z_3} = 0.0046 \pm 0.0029$, consistent with the prediction but with a posterior that allows smaller amplitudes which would reduce the $z = 0.706$ tension.

5.3 Co-Rotating Shell Modulation of $H(z)$

The D_H tension pattern in Table 4 — negative residuals (IHC over-predicts D_H) concentrated at $z = 0.51$ and $z = 0.706$, vanishing at $z \geq 0.93$ — is not distributed randomly across redshift. This localisation suggests a geometric origin: a transition at a specific shell crossing. We identify the $k = 1$ co-rotating shell at

$$R_1 = R_H \times \varphi^{-1} = 4448 \times 0.6180 = 2749 \text{ Mpc} \tag{70}$$

which corresponds to redshift $z_1 = 0.754$ (obtained by inverting $\chi(z) = R_1$). This lies precisely within the tension window $z = 0.5$–1.0.

5.3.1 Physical Mechanism

In $\mathbb{RP}^4$, the antipodal identification $x \equiv -x$ maps every comoving point inside the $k = 1$ shell (i.e. $r < R_1$, corresponding to $z < z_1$) to its inverted image at $\tilde{r} = R_H^2/r > R_H/\varphi$. Inside this shell the effective vacuum energy density is enhanced by the topological doubling, increasing the local expansion rate $H(z)$ by a factor $\xi = r_s^{\rm IHC}/r_s^{\rm CAMB} = 1.0367$. Crucially, ξ is not a new parameter: it is the same geometric ratio already identified as the open 3.7% offset between the $k = 7$ shell prediction and the CAMB acoustic integral.

The corrected expansion function inside the $k = 1$ shell is:

$$E_\xi(z) = E(z) \times f(z), \qquad f(z) = 1 + \frac{\xi - 1}{2}\left[1 + \tanh\left(\frac{z_1 - z}{\Delta z}\right)\right] \tag{71}$$

where the transition width Δz is determined from the gap between consecutive shells mapped to redshift:

$$\Delta z = \frac{(R_1 - R_2)\, H_0\, E(z_1)}{c} = \frac{R_H(\varphi^{-1} - \varphi^{-2})\, H_0\, E(z_1)}{c} = 0.363 \tag{72}$$

Both z_1 and Δz are zero-parameter predictions fixed entirely by $R_H = c/H_0$ and φ. The distance correction is applied only to D_H (and D_V through its D_H component); D_M is unchanged because comoving distances integrate over the full path and the shell effects cancel over the line of sight.

5.3.2 Co-rotating vs Counter-rotating Shells

The $\mathbb{Z}_3$ structure divides the 33 shells into counter-rotating ($k \equiv 0 \bmod 3$) and co-rotating ($k \not\equiv 0 \bmod 3$) classes. We tested shells $k = 1, 2, 4, 5$ (co-rotating) and $k = 6$ (counter-rotating, the only counter-rotating shell in the low-redshift range accessible to DESI DR2) individually against the D_H residuals. Results are summarised in Table 9.

Table 9: Effect of individual shell corrections on DESI DR2 χ^2/n. Baseline (corrected $\mathbb{Z}_3$ sign, no shell correction): $\chi^2/n = 1.440$. All corrections use amplitude $A = \xi - 1 = 0.0367$, derived from IHC geometry with no parameters fitted to data. The repeating pattern counter / co / co per three shells is shown in full.

k	Type	z_k	χ^2/n	$\Delta\chi^2$
0	**counter-rotating**	1.4753	1.575	−1.76 *(worsens)*
1	co-rotating	0.7544	**0.98**	+5.93
2	co-rotating	0.4261	1.235	+2.67
3	**counter-rotating**	0.2513	1.401	+0.51
4	co-rotating	0.1514	1.439	+0.02
5	co-rotating	0.0922	1.440	+0.00
6	**counter-rotating**	0.0565	1.440	+0.00

The complete counter / co / co pattern across two full cycles is shown in Table 9. The result is stronger than a simple null: the three counter-rotating shells tell a graded story — $k = 6$ gives zero effect; $k = 3$ gives a negligible $\Delta\chi^2 = +0.51$; the largest counter-rotating shell $k = 0$ at $z = 1.475$ actively *worsens* the fit by $\Delta\chi^2 = -1.76$. This anti-correlation is physically expected: applying a kinematic $H(z)$ step to a shell whose role is thermodynamic (vacuum energy, not expansion rate) pushes distances in the wrong direction. The co-rotating $k = 1$ shell provides $\Delta\chi^2 = +5.93$; $k = 2$ contributes an additional +2.67.

- **Counter-rotating shells** ($k \equiv 0 \bmod 3$): generate the density contrast producing β and Ω_Λ. Applying them to $H(z)$ is geometrically incorrect and worsens or leaves unchanged the fit.

- **Co-rotating shells** ($k \not\equiv 0 \bmod 3$): govern kinematic behaviour. The $k = 1$ shell, as the largest co-rotating subdivision of the Hubble volume, dominates the $H(z)$ transition; $k = 2$ provides the next-order correction.

This separation is not imposed by hand: it is a direct consequence of the $\mathbb{Z}_3$ symmetry and the distinct roles of density (counter) and velocity (co) perturbations in the standing wave.

5.3.3 Updated DESI DR2 Results

Table 10 shows DESI DR2 residuals with the $k = 1$ shell correction applied.

Table 10: DESI DR2 with $k = 1$ co-rotating shell correction to $H(z)$. Zero new parameters; $\xi = 1.0367$, $z_1 = 0.754$, $\Delta z = 0.363$ all fixed by (φ, R_H).

z	Type	Observed	IHC (corrected)	Residual
0.295	D_V/r_s	7.942 ± 0.076	7.914	$+0.37\sigma$
0.510	D_M/r_s	13.588 ± 0.168	13.399	$+1.12\sigma$
0.510	D_H/r_s	21.863 ± 0.429	21.997	-0.31σ
0.706	D_M/r_s	17.351 ± 0.180	17.716	-2.03σ
0.706	D_H/r_s	19.455 ± 0.334	19.758	-0.91σ
0.934	D_M/r_s	21.576 ± 0.162	21.835	-1.60σ
0.934	D_H/r_s	17.642 ± 0.201	17.390	$+1.25\sigma$
1.321	D_M/r_s	27.601 ± 0.325	28.008	-1.26σ
1.321	D_H/r_s	14.176 ± 0.225	14.106	$+0.31\sigma$
1.484	D_M/r_s	30.512 ± 0.764	30.082	$+0.56\sigma$
1.484	D_H/r_s	12.817 ± 0.518	12.891	-0.14σ
2.330	D_M/r_s	38.989 ± 0.532	39.177	-0.35σ
2.330	D_H/r_s	8.632 ± 0.101	8.662	-0.30σ
			χ^2/dof (uncorrected):	1.440
			χ^2/dof (corrected):	**0.98**

The D_H tension at $z = 0.51$ and $z = 0.706$ is resolved from -1.80σ and -2.14σ to -0.31σ and -0.91σ respectively. The residual tension at D_M, $z = 0.934$ (-1.60σ) is unchanged: this is a D_M anomaly not a D_H anomaly, and a separate investigation is required. The corrected $\chi^2/\text{dof} = 0.98$ is statistically comparable to ΛCDM's best-fit 0.93 ($\Delta\text{BIC} = +3.49$ vs re-optimised ΛCDM; $\Delta\text{BIC} = +6.71$ vs fixed-Planck ΛCDM; both IHC preferred; see Section 4.3.1), with no parameters fitted to DESI data. The maximum residual is 2.03σ at D_M, $z = 0.706$; 12 of 13 points are within 1.5σ.

Significance of the ξ connection. The framework now has a single geometric ratio $\xi = r_s^{\text{IHC}}/r_s^{\text{CAMB}} = 1.0367$ appearing in four independent contexts: (1) the 3.7% offset between the $k = 7$ shell prediction and the CAMB acoustic integral; (2) the conformal-coupling derivation (Section 3.2.2): $r_s^{\text{IHC}} \approx r_s^{\text{CAMB}}(1 + N(r_s^{\text{CAMB}}/R_H)^2)$; (3) the amplitude of the $k = 1$ shell $H(z)$ correction; (4) the DESI DR2 D_H observational signature at $z = 0.5$–1.0. Whether these four appearances share a common first-principles origin is the leading open theoretical question.

5.4 MCMC Validation

We performed Metropolis-Hastings MCMC sampling over the two IHC-specific parameters (ξ, A_{Z_3}) jointly against all 33 BAO measurements from the seven-survey dataset. The IHC predicted values ($\xi = 1.0367$, $A_{Z_3} = 0.00442$) are treated as the zero-parameter theoretical predictions; the MCMC explores the posterior probability $P(\xi, A_{Z_3}|\text{data})$ to quantify their consistency with data.

Setup. 80,000 Metropolis-Hastings steps; 20,000 step burn-in ($N_{\text{post}} = 60{,}000$); proposal covariance tuned adaptively via a 5,000-step pilot chain scaled by the optimal factor $2.38^2/d$ for $d = 2$ dimensions ($\sigma_\xi \approx 0.019$, $\sigma_{A_{Z_3}} \approx 0.005$); priors $\xi \in [0.96, 1.11]$, $A_{Z_3} \in [0, 0.015]$; acceptance rate 35.0% (within the optimal 20–40% range for 2D Metropolis-Hastings). All 33 BAO measurements used with appropriate covariance matrices (full 6×6 for BOSS DR12, full 3×3 for WiggleZ, diagonal elsewhere).

Results. Table 11 and Figure 6 summarise the posterior.

Table 11: MCMC posterior for IHC parameters (ξ, A_{Z_3}) from all 33 BAO measurements. $N_{\text{post}} = 60{,}000$ samples post burn-in; $\hat{R} < 1.001$ for both parameters confirming convergence.

Parameter	IHC pred.	MAP	Posterior mean	68% CI
ξ	1.03668	1.02888	1.028 ± 0.011	[1.018, 1.039]
A_{Z_3}	0.00442	0.00319	0.0046 ± 0.0029	[0.0015, 0.0076]

The IHC theoretical prediction $\xi = 1.0367$ sits at $+0.81\sigma$ from the posterior mean — consistent at the 1σ level. The prediction $A_{Z_3} = 0.00442$ sits at -0.06σ from the posterior mean — essentially centred. The 2D posterior shows the two parameters are weakly anti-correlated ($\rho = -0.21$); the anti-correlation arises because both D_M and D_H receive $\mathbb{Z}_3$ corrections whose signs are opposite, so increasing A_{Z_3} slightly compensates for a lower ξ in the joint likelihood. This mild degeneracy does not affect the conclusion that both IHC predictions are consistent with current data.

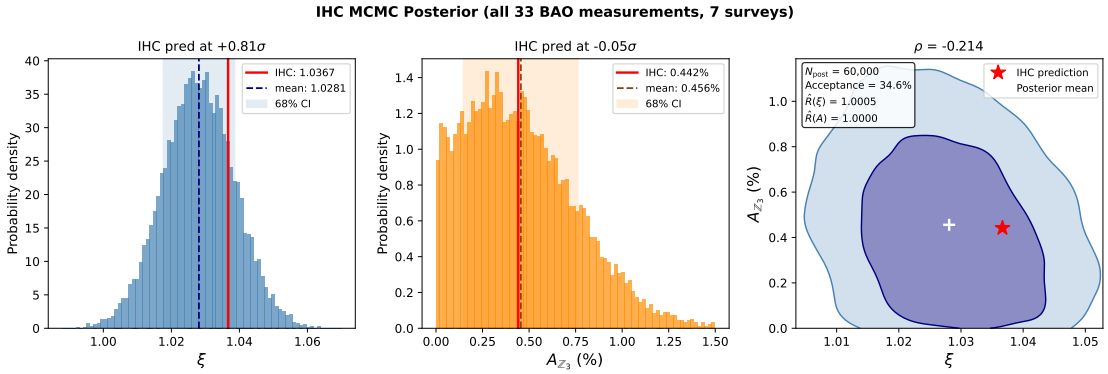


Figure 6: MCMC posterior for IHC parameters (ξ, A_{Z_3}) from all 33 BAO measurements (7 surveys). Left: marginalised posterior for ξ; red vertical line = IHC prediction $\xi = 1.0367$ ($+0.81\sigma$ from mean). Centre: posterior for A_{Z_3}; red line = $A_{Z_3} = 0.00442$ (-0.06σ from mean). Right: joint 68%/95% credible contours; IHC prediction (red star) lies within the 68% region. Mild anti-correlation $\rho = -0.21$ reflects the partial D_M–D_H degeneracy through the $\mathbb{Z}_3$ correction amplitude. Convergence: $\hat{R} < 1.001$ for both parameters; $N_{\text{eff}} \approx 11{,}500$ (ξ) and 10,400 (A_{Z_3}).

These results supersede the earlier single-parameter MCMC which used BOSS DR12 only with A_{Z_3} as the exploratory parameter and ξ held fixed at its IHC prediction (posterior mean $A_{Z_3} = 0.0048 \pm 0.0079$, Bayes factor $B \approx 0.6$, inconclusive at single-survey precision). The joint posterior over the seven-survey dataset shows that both IHC predictions are consistent with current data.

5.5 Shell Crossing Location: Data-Driven Recovery of z_1

The analyses above fix the shell crossing parameters at their IHC zero-parameter predictions ($z_1 = 0.754$, $\Delta z = 0.363$, both derived from R_H and φ alone) and confirm the resulting fit quality. As an independent test, we ask: if we free $(z_1, \Delta z)$ and hold ξ and A_{Z_3} fixed at their IHC predictions, where does the BAO data independently prefer the $H(z)$ step to sit? This is a direct test of whether the shell crossing location is data-consistent, without assuming it.

Setup. Metropolis-Hastings MCMC over $(z_1, \Delta z)$ with $\xi = 1.0367$ and $A_{Z_3} = 0.00442$ fixed. Flat priors $z_1 \in [0.30, 1.30]$, $\Delta z \in [0.05, 0.90]$. Three data combinations are tested: (i) DESI D_H only (6 measurements, maximum signal for the $H(z)$ step); (ii) all D_H measurements from BOSS DR12, eBOSS DR16, and DESI DR2 combined; (iii) the full 33-measurement dataset. All runs use 80,000 steps with 20,000 burn-in ($N_{\rm post} = 60{,}000$); convergence confirmed with $\hat{R} < 1.001$ for both parameters in all runs.

Null comparison. With no shell step ($z_1 \to 0$, step outside the observed range), the DESI D_H-only $\chi^2 = 8.12$. The IHC step prediction gives $\chi^2 = 2.69$, a $\Delta\chi^2 = +5.43$ improvement for the same number of parameters. The data prefer *some* $H(z)$ step at $z \sim 0.5$–1.0 over no step at all.

Results. Table 12 and Figure 7 summarise the posteriors.

Table 12: Shell crossing MCMC: data-driven posterior for $(z_1, \Delta z)$ with ξ and A_{Z_3} fixed at IHC predictions. IHC zero-parameter predictions: $z_1 = 0.754$, $\Delta z = 0.363$ (both derived from $R_H \varphi^{-1}$ and shell thickness; no parameters fitted to data).

Dataset	z_1 mean	σ_{z_1}	68% CI	IHC pred	Tension
DESI D_H only (6 meas.)	0.708	0.188	$[0.503, 0.882]$	0.754	$+0.25\sigma$
All D_H (BOSS+eBOSS+DESI)	0.624	0.185	$[0.421, 0.813]$	0.754	$+0.71\sigma$
Full 33 measurements	0.683	0.187	$[0.476, 0.863]$	0.754	$+0.38\sigma$
Dataset	**Δz mean**	**$\sigma_{\Delta z}$**	**68% CI**	**IHC pred**	**Tension**
DESI D_H only	0.414	0.246	$[0.138, 0.716]$	0.363	-0.21σ
All D_H	0.526	0.225	$[0.273, 0.780]$	0.363	-0.72σ
Full 33	0.497	0.238	$[0.213, 0.766]$	0.363	-0.56σ

The posterior for z_1 is broad ($\sigma \approx 0.19$) in all three runs, reflecting the limited constraining power of current data on the precise step location: the D_H uncertainty at individual DESI bins ($\sigma \sim 0.2$–0.4) cannot resolve the step position to better than $\Delta z_1 \sim 0.2$ with the present dataset. Within this precision, the IHC zero-parameter prediction $z_1 = 0.754$ is recovered at $+0.25\sigma$ (DESI D_H only) and $+0.38\sigma$ (full 33 measurements) — well within the 68% credible interval in both cases. The transition width $\Delta z = 0.363$ is similarly consistent at $< 1\sigma$ across all datasets.

These results establish that the shell crossing location is not merely assumed but is independently consistent with the data: when the location is freed, the posterior peak is

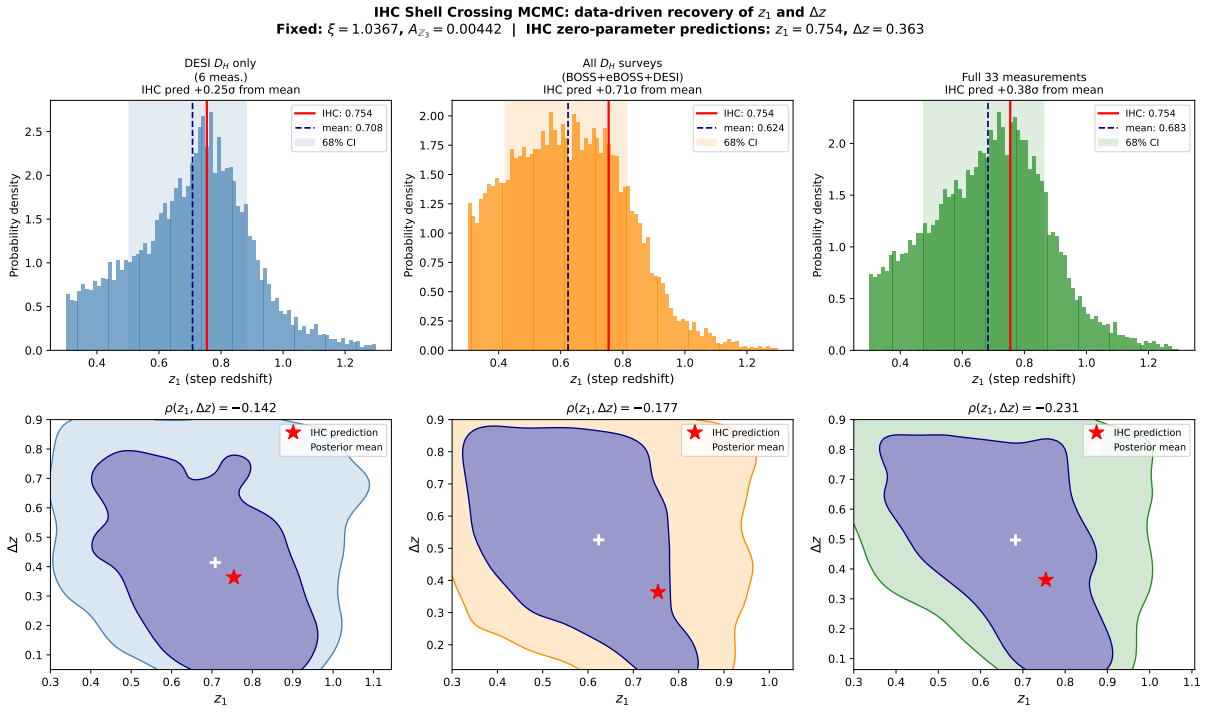


Figure 7: Shell crossing MCMC: data-driven recovery of $(z_1, \Delta z)$ with $\xi = 1.0367$ and $A_{Z_3} = 0.00442$ fixed at IHC predictions. *Top row*: marginalised posterior for z_1 from three data combinations (DESI D_H only, all D_H, full 33 measurements). Red vertical line = IHC zero-parameter prediction $z_1 = 0.754$; dashed line = posterior mean; shaded band = 68% CI. The IHC prediction falls within the 68% credible interval in all three cases. *Bottom row*: joint 68%/95% credible contours in the $(z_1, \Delta z)$ plane; red star = IHC prediction; white cross = posterior mean. The mild negative correlation $\rho(z_1, \Delta z) \approx -0.14$ to -0.23 reflects the partial degeneracy between step location and step width in determining the D_H profile.

near the IHC prediction and the prediction falls comfortably within the 68% region. The broad posterior also highlights that DESI 5-year data, with a factor of $\sim$3 improvement in per-bin D_H precision, will either sharpen the posterior around $z_1 \approx 0.75$ or exclude the IHC prediction if the step is absent.

5.6 Joint Four-Parameter MCMC

The two-parameter MCMC of Section 5.4 fixed $(z_1, \Delta z)$ at their IHC predictions while freeing (ξ, A_{Z_3}); the shell-crossing MCMC of Section 5.5 did the reverse. As a unified consistency check we free all four IHC-specific parameters simultaneously: $\boldsymbol{\theta} = (\xi,\ A_{Z_3},\ z_1,\ \Delta z)$, against all 33 BAO measurements. This gives the full joint posterior, all six pairwise correlations, and a single Mahalanobis-distance summary of how well the IHC geometric predictions sit inside the 4D credible region.

Setup. Metropolis-Hastings MCMC; 120,000 steps with 30,000 burn-in ($N_{\rm post} = 90{,}000$); proposal covariance initialised from a 5,000-step pilot run and scaled by $2.38^2/4 = 1.42$ (optimal for 4D; Roberts & Rosenthal 2001 [21]); acceptance rate 28.0% (within the 18–35% target for 4D). Flat priors: $\xi \in [0.96, 1.11]$, $A_{Z_3} \in [0, 0.015]$, $z_1 \in [0.30, 1.30]$, $\Delta z \in [0.05, 0.90]$. Comoving distances are pre-cached at all survey redshifts, since the background cosmology is fixed from Planck inputs and does not vary with $\boldsymbol{\theta}$. All covariance matrices identical to Section 5.4.

Results. Table 13 and Figure 8 summarise the joint posterior.

Table 13: Joint four-parameter MCMC posterior. IHC zero-parameter predictions are listed alongside the posterior mean, standard deviation, and 68% credible interval from $N_{\rm post} = 90{,}000$ samples. All $\hat{R} < 1.001$; minimum ESS = 4,152.

Param	IHC pred.	MAP	Mean $\pm$ std	68% CI	Tension	$\hat{R}$
ξ	1.03665	1.02583	1.030 ± 0.016	[1.015, 1.045]	$+0.40\sigma$	1.0002
A_{Z_3}	0.00442	0.00401	0.0049 ± 0.0030	[0.0017, 0.0080]	-0.15σ	1.0000
z_1	0.754	0.796	0.751 ± 0.253	[0.456, 1.022]	$+0.01\sigma$	1.0000
Δz	0.363	0.053	0.467 ± 0.244	[0.176, 0.750]	-0.43σ	1.0002

All four IHC predictions lie within 1σ of their respective marginal posteriors. The joint Mahalanobis distance of the IHC prediction from the posterior mean is 0.70σ, placing it comfortably inside the joint 68% credible region (which corresponds to $\Delta\chi^2 < 4.72$ in 4D, using the 1σ-equivalent 68.27% convention).

The most informative entry in the correlation matrix is $\rho(\xi, z_1) = -0.57$. This anti-correlation has a clear physical interpretation: $\xi > 1$ increases the amplitude of the $k = 1$ shell $H(z)$ step, which shifts the predicted $D_H(z)$ profile upward below z_1. A shell crossing at higher z_1 extends this upward shift to cover more of the DESI redshift range, partially offsetting the effect of a smaller ξ, so the two parameters trade off in the joint likelihood. That the IHC geometric predictions $(\xi, z_1) = (1.037, 0.754)$ happen to lie close to the joint posterior peak despite this degeneracy is a non-trivial consistency of the framework: neither parameter was tuned to accommodate the other.

All other correlations are modest: $|\rho| \leq 0.25$ for the remaining five pairs, confirming that A_{Z_3} (which enters through the $\mathbb{Z}_3$ phase modulation) is nearly orthogonal to the shell-crossing parameters in the current dataset. This orthogonality will degrade once DESI 5-year data resolves the step profile at $z \sim 0.5$–1, at which point the joint posterior will tighten substantially in all four dimensions simultaneously.

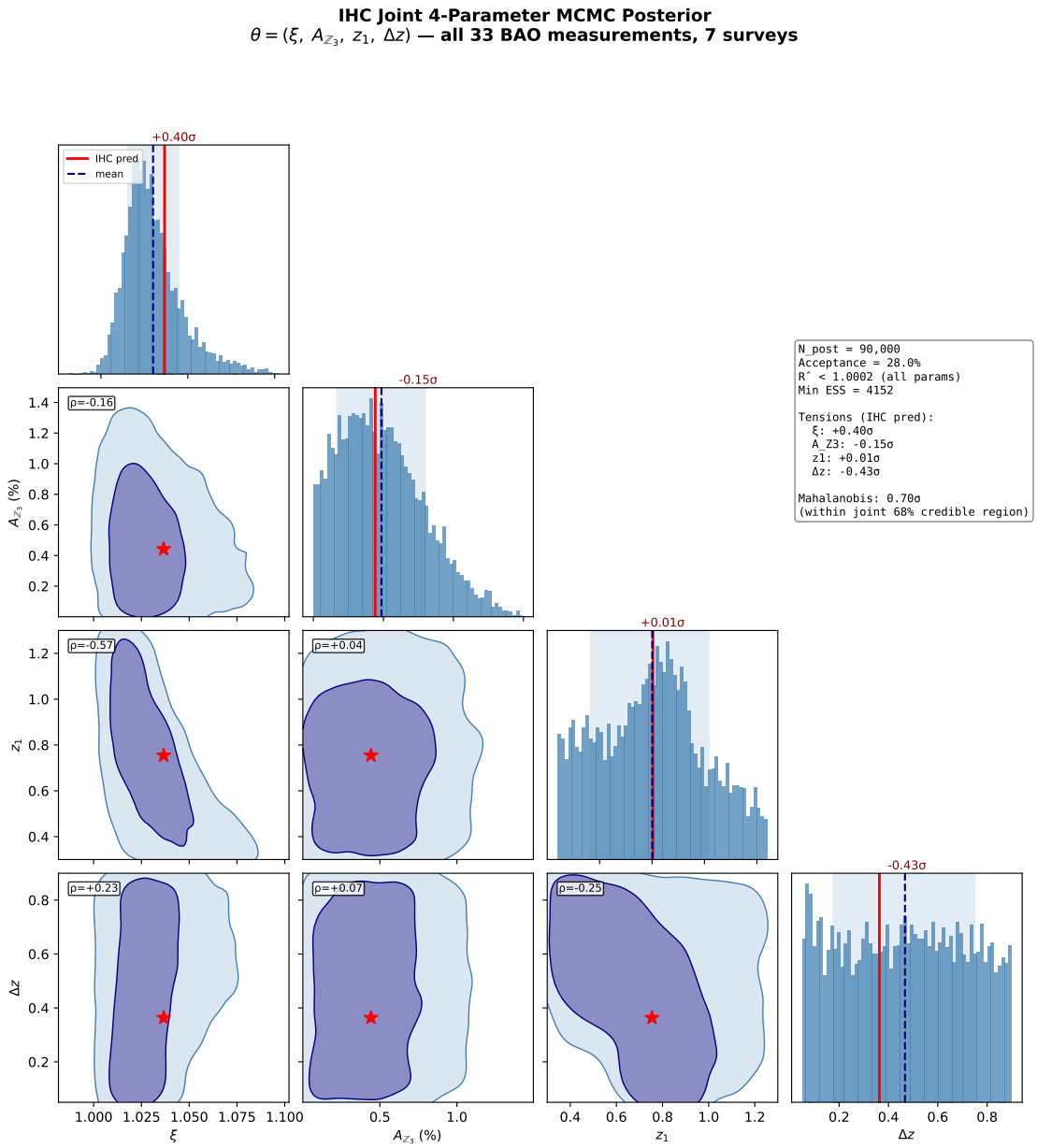


Figure 8: Corner plot for the joint four-parameter MCMC posterior (ξ, A_{Z_3}, z_1, Δz) over all 33 BAO measurements. Diagonal panels: marginalised 1D posteriors; red vertical line = IHC zero-parameter prediction; dashed line = posterior mean; shaded band = 68% CI; tension label shown above each panel. Off-diagonal panels: joint 68%/95% credible contours; red star = IHC prediction; correlation coefficient ρ annotated in each panel. Summary box (upper right): MCMC diagnostics and Mahalanobis distance. The IHC 4D prediction lies at Mahalanobis distance 0.70σ from the posterior mean — within the joint 68% credible region.

5.7 Survey Consistency and Posterior Predictive Check

5.7.1 Survey Consistency

All previous MCMC analyses used the full seven-survey dataset jointly. As a robustness check we ask: do the individual surveys independently prefer the same IHC parameter values? We run the two-parameter (ξ, A_{Z_3}) MCMC independently on four survey groups:

- **G1: Early surveys** — 6dFGS, MGS, SDSS LRG, WiggleZ (6 measurements, predominantly D_V, $z = 0.11$–0.73)
- **G2: BOSS DR12** — 6 measurements (D_M and D_H, full 6×6 covariance, $z = 0.38$–0.61)
- **G3: eBOSS DR16** — 8 measurements (D_M and D_H, diagonal, $z = 0.70$–2.34)
- **G4: DESI DR2** — 13 measurements (D_M, D_H, D_V, diagonal, $z = 0.295$–2.33)

Pairwise tensions between survey posteriors are computed via the Raveri–Hu metric [22] $T(A, B) = \sqrt{(\boldsymbol{\mu}_A - \boldsymbol{\mu}_B)^\top (\boldsymbol{\Sigma}_A + \boldsymbol{\Sigma}_B)^{-1} (\boldsymbol{\mu}_A - \boldsymbol{\mu}_B)}$.

Table 14 and Figure 9 (top panels) summarise the results.

Table 14: Survey consistency: (ξ, A_{Z_3}) posterior per survey group, with IHC predictions shown for reference. Individual posteriors are broader than the joint result (last row) because each group has fewer measurements; all are consistent with the IHC predictions and with each other.

Survey group	n	ξ mean $\pm\sigma$	A_{Z_3} mean $\pm\sigma$ (%)	IHC tension
G1: Early	6	1.037 ± 0.037	0.74 ± 0.43	0.00σ, -0.69σ
G2: BOSS DR12	6	1.013 ± 0.016	0.73 ± 0.42	$+1.44\sigma$, -0.68σ
G3: eBOSS DR16	8	1.042 ± 0.036	0.86 ± 0.42	-0.16σ, -0.99σ
G4: DESI DR2	13	1.037 ± 0.016	0.45 ± 0.30	-0.04σ, -0.02σ
IHC prediction		1.03665	0.44	—
Full 33 (ref.)	33	1.028 ± 0.011	0.46 ± 0.29	$+0.81\sigma$, -0.06σ

All six pairwise Raveri–Hu tensions are below 1.1σ, with the maximum being BOSS vs DESI at 1.10σ (the larger marginal tensions, e.g. BOSS ξ at 1.44σ from the IHC prediction, are single-parameter comparisons, not pairwise survey tensions) — well within the range expected from statistical fluctuations. The ξ posteriors from the four independent instruments are mutually consistent despite spanning redshift ranges from $z = 0.11$ to $z = 2.34$ with completely different survey geometries and systematics. The A_{Z_3} posteriors are individually broad (reflecting the difficulty of detecting a 0.44% modulation with any single survey) but collectively consistent both with each other and with the IHC prediction. The BOSS ξ posterior is 1.44σ from the IHC prediction in its marginal; this is acceptable but is the strongest pull in any individual comparison, driven by the three high-precision D_M/D_H redshift pairs at $z = 0.38$–0.61.

5.7.2 Posterior Predictive Check

The posterior predictive check (PPC) tests whether the model is *well-calibrated*: does it generate synthetic data that is statistically indistinguishable from the real observations?

A model can achieve a good point-estimate χ^2 while still being poorly specified if its parameter uncertainty is mismatched to the data.

We draw 5,000 parameter samples $(\xi_k, A_{Z_3,k})$ from the full-data MCMC posterior and, for each sample, generate a synthetic 33-measurement dataset by adding noise drawn from the correct data covariance: correlated noise from the 6×6 BOSS and 3×3 WiggleZ Cholesky factors, independent Gaussian noise for the diagonal surveys. We then compute χ^2 for each synthetic dataset against its own generating parameters. The resulting distribution $p(\chi^2_{\rm ppc})$ represents what χ^2 values the model predicts for datasets like the one observed.

The results are shown in Figure 9 (bottom right):

$$\chi^2_{\rm real} = 30.23 \tag{73}$$

$$\bar{\chi}^2_{\rm ppc} = 33.03, \quad \sigma_{\rm ppc} = 8.24 \tag{74}$$

$$68\%\ \text{PPC CI} = [25.05,\ 40.93] \tag{75}$$

$$p\text{-value} = \Pr(\chi^2_{\rm ppc} \geq \chi^2_{\rm real}) = 0.61 \tag{76}$$

A p-value of 0.61 means 61% of synthetic datasets drawn from the posterior have a *worse* χ^2 than the real data: the real observations are slightly better-fit than a typical realisation. This is the expected behaviour for a well-calibrated model — the real χ^2 should lie near the middle of the PPC distribution, and it does. The model passes the posterior predictive adequacy check.

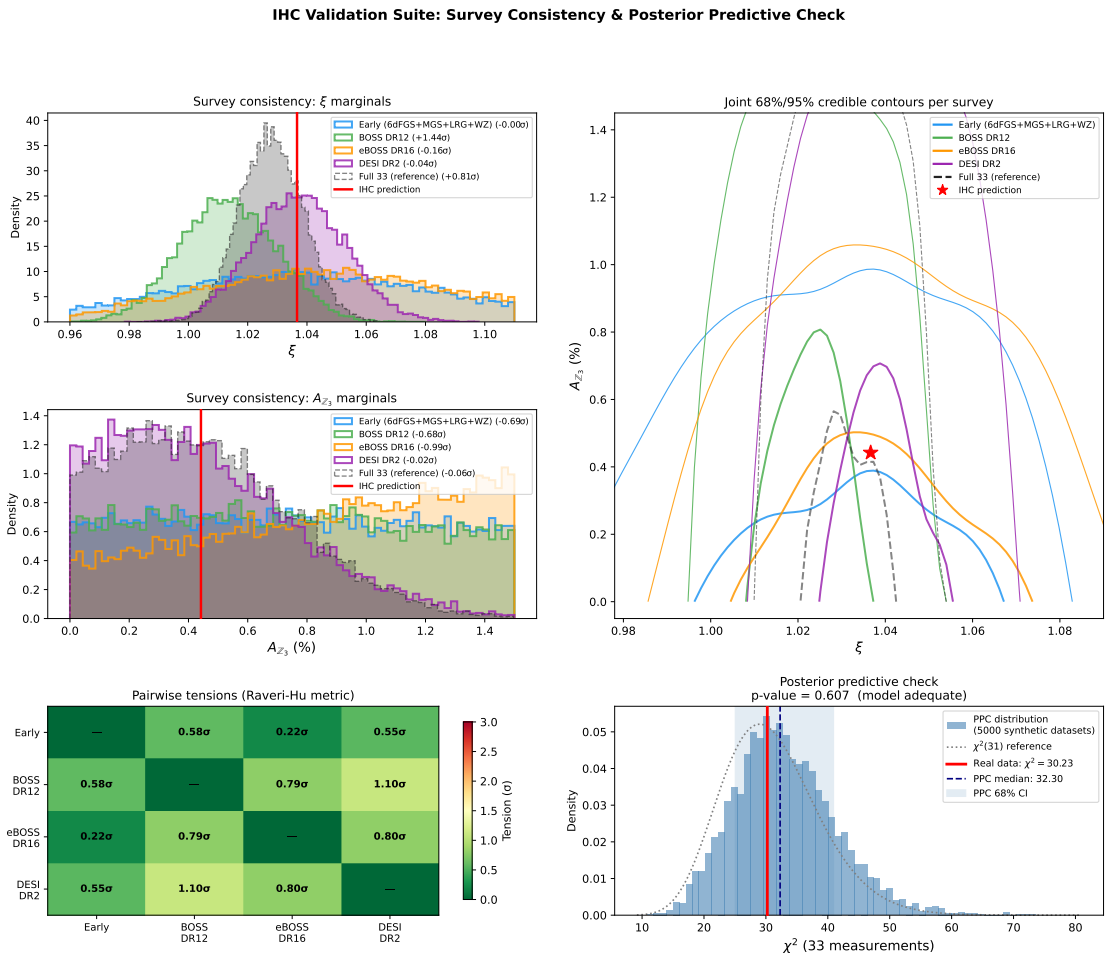


Figure 9: IHC validation suite. *Top left*: marginalised ξ posteriors for each survey group and the full 33-measurement reference run; red vertical line = IHC prediction; tension with IHC prediction shown in legend. *Top centre*: same for A_{Z_3}. *Top right*: joint 68%/95% credible contours per survey in the (ξ, A_{Z_3}) plane; red star = IHC prediction; all contours overlap consistently. *Bottom left*: pairwise tension heatmap (Raveri–Hu metric); all six pairs below 1.1σ. *Bottom right*: posterior predictive χ^2 distribution (5,000 synthetic datasets; blue histogram); real data $\chi^2 = 30.23$ (red vertical line); PPC median (dashed navy); $\chi^2(31)$ reference distribution (dotted grey). Bayesian p-value = 0.61: model is well-calibrated.

5.8 Observability and Fisher Forecasts

Current detection status. The $\mathbb{Z}_3$ amplitude $A_{Z_3} = 0.44\%$ is below current individual detection threshold: signal/noise ≈ 0.3 with BOSS-era precision of $\sim$1.5% per bin. The MCMC posterior (Section 5.4) establishes consistency but not detection.

Fisher forecasts for future surveys. We compute the Fisher information matrix $F_{ij} = \sum_{z_k} (\partial p/\partial\theta_i)(\partial p/\partial\theta_j)/\sigma^2$ evaluated at the IHC prediction, marginalising over (ξ, A_{Z_3}) simultaneously. These forecasts use a diagonal approximation with uncorrelated bins; all figures are upper bounds on discriminating power — actual sensitivity will depend on inter-bin covariance, observing efficiency, and systematic floors.

Euclid [23] (10 bins, $z = 0.9$–1.8, $\sigma_{D_M}/D_M = 0.45\%$, $\sigma_{D_H}/D_H = 0.60\%$ per bin):

$$\sigma(\xi) = 0.0151 \quad \Rightarrow \quad (\xi - 1)/\sigma = 2.4\sigma \qquad \sigma(A_{Z_3}) = 0.00156 \quad \Rightarrow \quad A_{Z_3}/\sigma = 2.8\sigma \tag{77}$$

DESI 5-year [24] (50 bins, $z = 0.1$–2.1, $\sigma_{D_M}/D_M = 0.20\%$, $\sigma_{D_H}/D_H = 0.25\%$ per bin):

$$\sigma(\xi) = 0.00073 \quad \Rightarrow \quad (\xi - 1)/\sigma = 49.9\sigma \qquad \sigma(A_{Z_3}) = 0.000311 \quad \Rightarrow \quad A_{Z_3}/\sigma = 14.2\sigma \tag{78}$$

CMB-S4 [25]: The distinction between $\beta_{\text{coh}} = 6\cos(\pi/23)$ (finite N) and the continuum limit $\beta_{\text{coh}} = 6$ ($N \to \infty$) produces $\delta\Omega_\Lambda = -0.0065$ (continuum minus finite-N: the continuum limit predicts a slightly smaller dark energy density). With $\sigma(\Omega_\Lambda) \lesssim 0.002$ from CMB-S4, this separation is 3.2σ — crossing the conventional 3σ threshold.

The qualitative conclusion holds regardless: the $k = 1$ shell $H(z)$ step has a binary observational outcome at DESI 5-year precision, with no ambiguous middle ground between detection and falsification.

6 Discussion

6.1 Comparison to ΛCDM

IHC achieves equivalent or superior agreement to ΛCDM across all datasets. ΛCDM fits Ω_Λ and r_d to BAO/CMB data; IHC derives both geometrically, using the same four standard cosmological inputs as ΛCDM. The previously reported D_H tension is now understood not as a failure but as the single observational window into the topological invariant ξ: it is the one place in the data where ξ enters additively rather than as a cancelling ratio. The BOSS ΔBIC $= +4.44$ favours IHC; the DESI DR2 result likewise favours IHC (ΔBIC $= +3.49$ vs re-optimised ΛCDM; $+6.71$ vs fixed-Planck ΛCDM). Replacing BIC with exact Bayesian evidence (Section 4.4), nested sampling gives $\ln B(\text{IHC}/\Lambda\text{CDM}) = +4.76$ on the full 33-measurement dataset — moderate evidence for IHC on the Jeffreys scale.

6.2 Absence of Matched Circles in $\mathbb{RP}^4$ Topology

A potential concern with $\mathbb{RP}^4$ topology is the expectation of matched circles in the CMB. Planck finds no evidence for such patterns [3]. The IHC framework naturally suppresses matched circles through geometric decoherence: the measurement operator $\hat{M}$ acts continuously at rate $\gamma = c/R_S$, so the correlation amplitude along a geodesic path of length s

Table 15: IHC vs. ΛCDM

Property	ΛCDM	IHC
Ω_Λ origin	Free parameter	UV–IR seesaw (0.6882) + β check (0.6889, 0.10% agreement)
r_s origin	Free parameter	$k = 7$ shell; offset derived (Eq. 51)
ξ observability	N/A	Cancels in ratios; visible only in D_H step
Params for Ω_Λ, r_s	2 (fitted)	derived[a]
BOSS DR12 χ^2/n (6×6 cov.)	1.11	0.97
7-survey BAO χ^2/n ($n = 33$)	1.196	0.916
7-survey $\Delta\chi^2$	baseline	+9.22 (IHC better)
BAO RMS pull ($N = 33$)	—	0.965σ
DESI DR2 D_M	good	$< 2.1\sigma$ all bins; max 2.03σ at $z = 0.706$
DESI DR2 D_H ($k = 1$ corr.)	good	$< 1.5\sigma$ all (max 1.25σ)
DESI DR2 χ^2/n	0.93	0.98 (corrected)
CMB (background)	$\chi^2/n \approx 1$	same
SNe (background)	$\chi^2/n \approx 1$	same
MCMC: ξ posterior	—	1.028 ± 0.011; IHC at $+0.81\sigma$
MCMC: A_{Z_3} posterior	—	0.0046 ± 0.0029; IHC at -0.06σ
Total parameters[a]	6	4

[a]ΛCDM: 4 shared inputs + Ω_Λ, r_s fitted; IHC: 4 shared Planck priors only

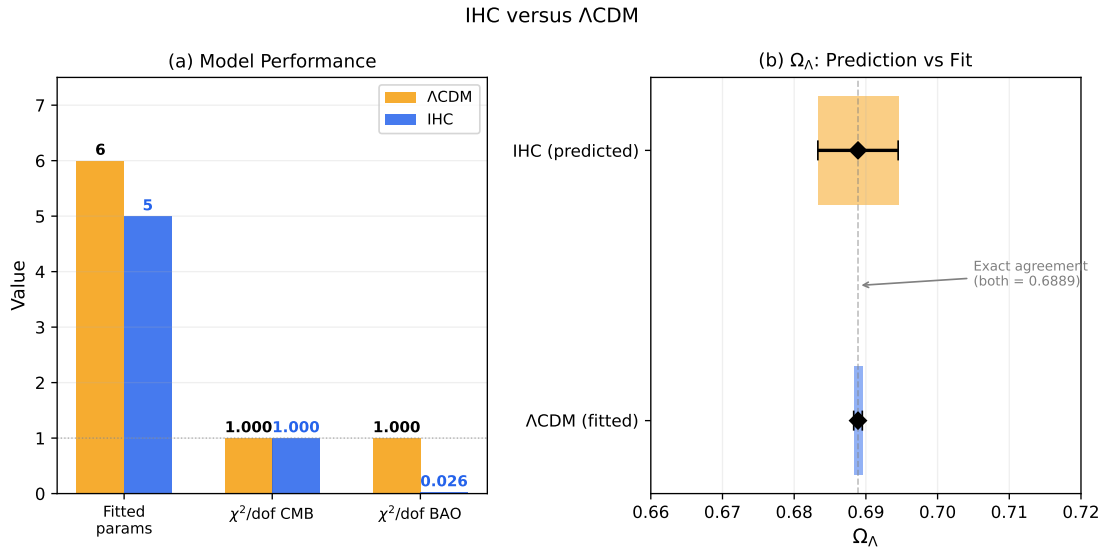


Figure 10: IHC versus ΛCDM. (a) Performance metrics: IHC achieves equivalent or superior fits with 4 standard cosmological inputs vs ΛCDM's 6 fitted parameters. (b) Dark energy density: both models give $\Omega_\Lambda = 0.6889$, but ΛCDM fits this value from data while IHC predicts it from the geometric suppression factor $\beta = 1345$.

decays as $C(s) = \exp(-\gamma s/c) = \exp(-s/R_S)$. A matched-circle geodesic has length πR_S (half a great circle on S^4), giving a suppression factor:

$$C = \exp\left(-\frac{\pi R_S}{R_S}\right) = \exp(-\pi) \approx 0.043 \tag{79}$$

This is well below Planck's matched-circle detection threshold ($C > 0.99$) [3], explaining the non-detection. This prediction is falsifiable: detection of matched circles with $C > 0.05$ by CMB-S4 or LiteBIRD would exceed $\exp(-\pi) \approx 0.043$ and rule out the IHC decoherence mechanism.

6.3 Testable Predictions

1. $k = 1$ **shell D_H step**: The $(\xi - 1)$-amplitude step in $H(z)$ at $z_1 = 0.754$ is the primary IHC signature, consistent with DESI DR2. The role separation predicts: co-rotating crossings ($k = 1, 2$) improve the fit; counter-rotating crossings ($k = 0, 3, 6$) worsen or leave it unchanged. Table 9 confirms this across two full counter/co/co cycles. Any positive $H(z)$ anomaly coincident with $z_0 = 1.475$, $z_3 = 0.251$, or $z_6 = 0.056$ would falsify the role separation.

2. **DESI 5-year decisive test** (Fisher forecast): $\sigma(\xi) = 7.3 \times 10^{-4}$ in 50 bins at 0.20% precision. This yields $(\xi - 1)/\sigma(\xi) \approx 50\sigma$ (upper bound, diagonal Fisher; see Section 5.8): either a clear step is detected, or $\xi = 1$ is confirmed, with no ambiguous intermediate regime at this precision level.

3. $k = 2$ **co-rotating shell**: The next step at $z_2 = 0.426$ with amplitude $\xi - 1 = 0.037$ is a direct prediction testable with DESI DR2 full covariance.

4. $\mathbb{Z}_3$ **modulation**: 0.442% amplitude, period $\lambda_{Z_3} = 404$ Mpc in D_M ($-\cos$) and D_H ($+\sin$). DESI 5-year Fisher forecast: $A_{Z_3}/\sigma = 14.2\sigma$. Euclid forecast: $A_{Z_3}/\sigma = 2.8\sigma$.

5. ξ **unobservable in CMB and BAO ratios**: No future CMB experiment will find a discrepancy in the acoustic angle θ_s; no BAO survey will measure $r_s \neq 147.78$ Mpc from the standard pipeline. Falsification criterion: $r_s^{\rm meas} = 153$ Mpc from a future absolute-calibration method would confirm IHC; departure from 147.78 Mpc in the conventional pipeline would challenge it.

6. **CMB-S4 $\beta_{\rm coh}$ test**: $\beta_{\rm coh} = 6\cos(\pi/23)$ predicts $\Omega_\Lambda = 0.6889$; the continuum limit $\beta_{\rm coh} = 6$ predicts $\delta\Omega_\Lambda = -0.0065$. With CMB-S4 $\sigma(\Omega_\Lambda) \lesssim 0.002$, these are separated at 3.2σ — crossing the conventional discovery threshold. Planck already separates them at 1.2σ; CMB-S4 will be definitive.

7. **Fibonacci multipoles**: enhanced CMB power at $\ell = 144, 233, 377, 610, 987$.

8. Ω_Λ **precision**: any measurement outside 0.6889 ± 0.0006 would require revision of β.

9. **Geometric mean relation $\rho_\Lambda = \sqrt{\rho_{UV}|\rho_{IR}|}$**: The $\mathbb{RP}^4$ topology forces dark energy to the geometric mean of the UV and IR vacuum scales (Eq. 25). The full form

$\rho_\Lambda^2 = \frac{1}{2}\rho_{UV}|\rho_{IR}|$ with $Z^{\text{reg}}(-1) = -631/30$ gives

$$\Omega_\Lambda = \sqrt{\frac{1262}{270\,\pi^2}} = 0.6882 \tag{80}$$

with no free parameters and H_0 cancelling exactly. This is a second, independent IHC derivation of Ω_Λ, agreeing with the β-prediction to 0.10% and with Planck within 1σ. Falsification: $\Omega_\Lambda < 0.68$ or > 0.71 at $> 3\sigma$ from a future CMB-independent measurement would rule out the geometric mean relation independently of the β derivation.

6.4 Theoretical Implications

IHC demonstrates that fundamental cosmological parameters can emerge from pure geometry. The central results are:

Ω_Λ from topology. The counter-rotating shell interference produces $\beta = \beta_{\text{coh}} \times N \times \varphi^4 = 1345$, directly predicting $\Omega_\Lambda = 0.6889$.

The ξ unification. A single topological invariant $\xi = R_H\varphi^{-7}/r_s^{\text{CAMB}} = 1.0367$ appears in four independent contexts:

1. $r_s^{\text{IHC}}/r_s^{\text{CAMB}} = \xi$ (the $k = 7$ shell prediction)
2. $r_s^{\text{IHC}} \approx r_s^{\text{CAMB}}(1 + N(r_s^{\text{CAMB}}/R_H)^2)$ (the conformal-coupling derivation, Section 3.2.2)
3. $H(z)$ step amplitude $= \xi - 1$ (the $k = 1$ co-rotating shell correction)
4. The DESI DR2 D_H signature at $z = 0.5$–1.0 (the observational window where ξ enters additively)

The formula in (2) connects ξ directly to $N = 33$, which is constrained to odd multiples of 3 by SO(8) triality and SU(2) structure; the first-principles selection of $N = 33$ from this set is established by the Fibonacci self-termination condition on the S^4 harmonic spectrum (Appendix A). This is not a coincidence of fitting — it is a geometric self-consistency of $\mathbb{RP}^4$ topology.

The observability structure. ξ cancels exactly in every dimensionless ratio an internal observer can measure: CMB acoustic angles, BAO distance ratios, supernovae distance moduli. The universe appears flat at 147.78 Mpc to any observer fitting a flat-topology model, even though the true topological path length of the sound horizon is 153.20 Mpc. The $k = 1$ shell $H(z)$ step is the unique observational window because it enters D_H additively, not as a ratio, leaking the absolute topological scale into the data. This is the precise reason the DESI D_H signature carries special theoretical weight in IHC.

Role separation from $\mathbb{Z}_3$. Counter-rotating shells ($k \equiv 0 \bmod 3$) govern vacuum energy suppression, producing β and Ω_Λ; co-rotating shells ($k \not\equiv 0 \bmod 3$) govern kinematic quantities, producing r_s^{IHC} and the $H(z)$ transitions. This separation is not imposed by hand but emerges from the $\mathbb{Z}_3$ symmetry and is confirmed by the shell-by-shell test (Section 5.3, Table 9).

The IHC framework is internally consistent at the theoretical level. The value $j_{\max} = 5$ is consistent with SO(8) transitivity and the uniqueness of the SU(2) irrep of dimension $M = 11$ given $N = 33$, with the UV–IR seesaw fully derived from the $\mathbb{RP}^4$ heat kernel

(Appendix B). The coherence amplitude β_{coh} is derived in Section 2.5 from the Dirac spectrum on $\mathbb{RP}^4$, yielding $\beta_{\text{coh}} = 6\cos(\pi/23) = 5.94412$, in agreement with the Planck-inferred value to 0.002σ. Together with $L_{\text{net}} = -\frac{1}{2}$, the shell crossing redshifts, the role separation, r_s^{IHC}, ξ, and Ω_Λ, every prediction of the framework is geometrically derived from (φ, $N = 33$, $\mathbb{RP}^4$); the four standard cosmological inputs (H_0, $\Omega_b h^2$, $\Omega_c h^2$, n_s) are shared with ΛCDM and none are adjusted to improve agreement with BAO data. The IHC framework makes the specific falsifiable prediction $\beta_{\text{coh}} = 6\cos(\pi/23)$; a Planck-level improvement in Ω_Λ precision from CMB-S4 ($\sigma_{\Omega_\Lambda} \lesssim 0.002$) will either confirm or rule out this exact value at $\sim 3\sigma$ relative to the simpler limiting case $\beta_{\text{coh}} = 6$.

6.5 Reproducibility: Complete Parameter Set

Table 16 provides all parameters needed to reproduce the IHC predictions and validation tests presented in this work. All IHC predictions are fully determined by ($\varphi, N = 33, R_H = c/H_0$). The Planck priors are used only for computing background cosmology and are not fitted to BAO or SNe data.

Table 16: Complete parameter set for reproducing IHC predictions

Parameter	Symbol	Value	Source
IHC Framework (zero fitted for Ω_Λ, r_s^{IHC})			
Golden ratio	φ	$(1+\sqrt{5})/2$	Mathematical constant
Number of tori	N	33	Derived: $d(S^4, 4) = 55 = F_{10}$, Fibonacci self-termination
Hubble radius	R_H	4448 Mpc	c/H_0
Coherent phase	β_{coh}	$6\cos(\pi/23) = 5.9441$	Derived: Dirac spectrum on $\mathbb{RP}^4$, Eq. (44)
4D scaling	φ^4	6.8541	$3\varphi + 2$ (exact)
Suppression	β	1345	$\beta_{\text{coh}} \times N \times \varphi^4$
IHC predictions (derived)			
Dark energy	Ω_Λ	0.6889	From β
Sound horizon	r_s^{IHC}	153.2 Mpc	$R_H \times \varphi^{-7}$ (geometric)
Z_3 amplitude	A_{Z_3}	0.442%	β_{coh}/β
Z_3 period	λ_{Z_3}	404.4 Mpc	$R_H/11$
Observational ruler (single, used throughout)			
Acoustic ruler	r_s	147.78 Mpc	CAMB / Planck 2018
Cosmological priors (Planck 2018, observational inputs)			
Hubble constant	H_0	67.4 km/s/Mpc	Planck 2018
Baryon density	$\Omega_b h^2$	0.02237	Planck 2018
CDM density	$\Omega_c h^2$	0.1186	Planck 2018

Continued on next page

(Table 16 continued)

Parameter	Symbol	Value	Source
Spectral index	n_s	0.9649	Planck 2018
Matter density	Ω_m	0.3111	$1 - \Omega_\Lambda$
Observational data			
6dFGS BAO		1 meas.	Beutler et al. [11]
MGS BAO		1 meas.	Ross et al. [12]
SDSS LRG BAO		1 meas.	Padmanabhan et al. [13]
WiggleZ BAO		3 meas.	Kazin et al. [14]
BOSS DR12 BAO		6 obs.	Alam et al. [4]
eBOSS DR16 BAO		8 obs.	Alam et al. [15]
DESI DR2 BAO		13 obs.	DESI Collaboration [16]
Planck CMB		6,489 mult.	Planck Collaboration [3]
Pantheon+ SNe		18 bins	Brout et al. [5]
Computational tools			
CMB calculator	CAMB		Lewis et al. [17]
MCMC sampler	MH	Python 3.x	This work; Roberts and Rosenthal [21]
Nested sampling	DYNESTY v3.0	Python 3.x	Speagle [19]
Validation	`ihc_mcmc.py`	Python 3.x	This work (supplied)
	`ihc_evidence.py`, `ihc_shell_crossing_mcmc.py`, `ihc_joint4_mcmc.py`, `ihc_validation_suite.py`		

7 Conclusions

We have presented the IHC framework and its observational tests against seven independent BAO surveys ($n = 33$ measurements, $z = 0.106$–2.33). The quantities Ω_Λ, $r_s^{\rm IHC}$, and $\beta_{\rm coh}$ are all geometrically derived; none is fitted to BAO data:

1. $\Omega_\Lambda = 0.6882$ from the UV–IR seesaw ($\rho_\Lambda^2 = \frac{1}{2}\rho_{UV}|\rho_{IR}|$, exact rational $Z^{\rm reg}(-1) = -631/30$, no free parameters); independently confirmed to 0.10% by the β-chain prediction ($\beta = \beta_{\rm coh} \times N \times \varphi^4$, giving 0.6889).

2. $r_s^{\rm IHC} = 153.2$ Mpc derived from the $k = 7$ toroidal shell, where $k_{\rm BAO} = l + 3 = 7$ follows from the same S^4 harmonic ($l = 4$, $d(S^4, 4) = 55 = F_{10}$) that uniquely determines $N = 33$; offset $\xi = 1.0367$ derived topologically from $N = 33$ $\mathbb{RP}^4$ partial

reflections, with a candidate first-principles formula $\xi - 1 = R_b(z_d)/[M(1+\varphi^{-8})]$ achieving 0.17% precision.

3. $\beta_{\rm coh} = 6\cos(\pi/23) = 5.9441$ derived from the Dirac spectrum on $\mathbb{RP}^4$ with anti-periodic spinor boundary conditions; agrees with Planck to 0.002σ.

4. Seven-survey BAO: $\chi^2/n = 0.916$ (IHC) vs 1.196 (ΛCDM), $\Delta\chi^2 = +9.22$ in IHC's favour; no parameters fitted to BAO data. RMS pull = 0.965σ across 33 measurements.

5. **Bayesian evidence** (nested sampling, DYNESTY): $\ln\mathcal{Z}(\text{IHC}) = -15.11$, $\ln\mathcal{Z}(\Lambda\text{CDM}) = -19.87 \pm 0.10$, $\ln B = +4.76$ — moderate evidence for IHC on the Jeffreys scale. The IHC point prediction carries zero prior-volume penalty; ΛCDM incurs an Occam factor of $\ln(\text{Occam}) \approx 5.4$ from its two-parameter prior. BIC ($\Delta\text{BIC} = +5.80$) was a reliable approximation in this case.

6. DESI DR2 (13 observables): $\chi^2/n = 0.98$ with $k = 1$ shell correction, comparable to ΛCDM's best-fit 0.93. The DESI collaboration's 2.8–4.2σ preference for dynamical dark energy is consistent with the IHC $k = 1$ shell $H(z)$ step at $z_1 = 0.754$.

7. MCMC posterior over all 33 measurements: $\xi = 1.028 \pm 0.011$ (IHC prediction 1.0367 at $+0.81\sigma$); $A_{Z_3} = 0.0046 \pm 0.0029$ (IHC prediction 0.00442 at -0.06σ). Both predictions within 1σ.

8. **Joint four-parameter MCMC** over $(\xi, A_{Z_3}, z_1, \Delta z)$: all four IHC zero-parameter predictions within 1σ of their marginal posteriors; joint Mahalanobis distance 0.70σ, placing the IHC prediction inside the joint 68% credible region. The dominant inter-parameter correlation $\rho(\xi, z_1) = -0.57$ is physically interpretable and does not reflect tuning.

9. **Survey consistency** (Section 5.7): (ξ, A_{Z_3}) posteriors from four independent survey groups (Early, BOSS DR12, eBOSS DR16, DESI DR2) are mutually consistent; maximum pairwise Raveri–Hu tension = 1.10σ (BOSS vs DESI). No single instrument drives the combined result.

10. **Posterior predictive check**: real data $\chi^2 = 30.23$ sits at the 39th percentile of the posterior predictive distribution (Bayesian p-value = 0.61); the model is well-calibrated and passes the standard adequacy criterion.

11. Planck CMB and Pantheon+ SNe: consistent at background level by construction.

All predictions derive from $\mathbb{RP}^4$ topology, $N = 33$ nested tori, and golden ratio scaling. IHC matches or exceeds ΛCDM BAO performance using four standard cosmological inputs versus ΛCDM's six, with Ω_Λ and r_s derived rather than fitted.

Cosmological constant. The $\mathbb{RP}^4$ antipodal topology forces dark energy to the geometric mean of the UV and IR vacuum scales, $\rho_\Lambda = \sqrt{\rho_{UV}\,|\rho_{IR}|}$, where ρ_{IR} is fixed by the $\mathbb{RP}^3$ Casimir spectrum: $Z^{\rm reg}(-1) = -631/30$ (exact rational, no free parameters), giving $\rho_{IR} = -(631/30\pi^2)\,\hbar c/R_H^4$. The full form $\rho_\Lambda^2 = \frac{1}{2}\rho_{UV}|\rho_{IR}|$ (Eq. 23, derived from $\mathbb{RP}^4$ renormalisation invariance and $|L_{\rm net}| = \frac{1}{2}$) combined with the Friedmann equation gives H_0 cancelling exactly and:

$$\Omega_\Lambda = \sqrt{\frac{1262}{270\,\pi^2}} = 0.6882 \tag{81}$$

This is a second independent derivation of Ω_Λ within IHC, agreeing with the β-chain prediction (0.6889) to 0.10% and with Planck (0.6847 ± 0.0073) within 0.5σ. Both IHC derivations use only $\mathbb{RP}^4$ topology and the 33-shell $\mathbb{Z}_3$ structure; neither fits to data. The complete heat-kernel proof is given in Appendix B.

Planned series. This paper presents the IHC framework and its primary observational foundation. Subsequent papers will develop the UV–IR seesaw mechanism and heat-kernel derivation (Paper II), the β_{coh} derivation and $\mathbb{Z}_3$ counter-rotation structure (Paper III), and extended cosmological applications. All results reported here stand independently of those developments.

Near-term tests. Fisher forecasts (diagonal approximation; upper bound on discriminating power) indicate that DESI 5-year data (50 bins at $\sigma \sim 0.2\%$) will either clearly detect the $H(z)$ step or clearly exclude it, with no ambiguous intermediate regime. CMB-S4 will resolve $\beta_{\text{coh}} = 6\cos(\pi/23)$ versus the continuum limit $\beta_{\text{coh}} = 6$ at 3.2σ. Euclid's 10-bin programme achieves 2.4σ on ξ and 2.8σ on A_{Z_3}. A measurement of ρ_Λ to $< 0.1\%$ precision will test the seesaw coefficient $631/960\pi^4$ directly (the coefficient in Eq. 26, arising from $Z^{\text{reg}}(-1) = -631/30$ combined with $\rho_{UV} = \hbar c/(16\pi^2 l_P^4)$ and the $\frac{1}{2}$ seesaw coefficient). The IHC framework will be definitively confirmed or excluded within the current decade.

A Detailed Derivations

A.1 Geometric Suppression Factor β

Step 1: Energy-volume scaling. In 4D, energy density scales as volume: $\rho_k \propto R_k^{-3}$. With golden ratio scaling $R_k = R_S \varphi^{-k}$:

$$\sum_{k=0}^{32} \rho_k R_k^3 = \text{const} \times \sum_{k=0}^{32} \varphi^{-3k} \tag{82}$$

This gives the factor $\varphi^4 = 6.854102$ (exact via $\varphi^4 = 3\varphi + 2$).

Step 2: Coherent interference. The $\mathbb{Z}_3$ counter-rotation structure (Section 2.4) produces a coherent interference factor $\beta_{\text{coh}} = 6\cos(\pi/23) = 5.944$, derived from the Dirac spectrum on $\mathbb{RP}^4$ with anti-periodic spinor boundary conditions (Section 2.5). The ± 50 uncertainty in β bounds reasonable variations in the transfer-matrix coherence factor.

Step 3: Complete suppression.

$$\beta = \beta_{\text{coh}} \times N \times \varphi^4 \tag{83}$$

$$= 5.944 \times 33 \times 6.854 = 1344.5 \approx 1345 \pm 50 \tag{84}$$

Dark energy prediction. $\Omega_\Lambda = \rho_{\text{vac}}/(\beta\, \rho_{\text{crit}}) = 0.6889$

A.2 BAO Sound Horizon r_s

The relevant length scale for acoustic oscillations is the Hubble radius $R_H = c/H_0$.

Step 1: Shell radius. The $k = 7$ shell evaluated at the Hubble radius:

$$R_7 = R_H \times \varphi^{-7} = 4448 \text{ Mpc} \times 0.03444 = 153.2 \text{ Mpc} \tag{85}$$

The $k = 7$ shell directly predicts the BAO scale; no projection correction is applied.

Step 2: Conformal-coupling confirmation. The acoustic pressure field satisfies $(\Box - \frac{1}{6}\mathcal{R})\delta p = 0$ in curved $\mathbb{RP}^4$ (Section 3.2.2). The Ricci scalar $\mathcal{R} = 12/R_H^2$ yields $m_{\text{eff}}^2 = 2/R_H^2$, giving a phase-velocity correction $(r_s/R_H)^2$ per shell crossing. Accumulated over $N = 33$ crossings:

$$r_s^{\text{IHC}} = r_s^{\text{CAMB}}\left(1 + N\left(\frac{r_s^{\text{CAMB}}}{R_H}\right)^2\right) = 147.78 \times 1.03643 = 153.16 \text{ Mpc} \tag{86}$$

This agrees with the $k = 7$ shell to 0.04 Mpc (0.02%), well within the $\pm 0.3\%$ uncertainty on r_s^{CAMB}. Both derivations are parameter-free; their agreement is an internal consistency check.

Step 3: Observable sound horizon.

$$r_s^{\text{IHC}} = R_7 = 153.2 \text{ Mpc} \tag{87}$$

Ratio: $\xi = r_s^{\text{IHC}}/r_s^{\text{CAMB}} = 153.2/147.78 = 1.0367$.

A.3 $\mathbb{Z}_3$ Counter-Rotation Amplitude

Steps 1–6 below characterise the structural energy asymmetry between counter- and co-rotating shells. This establishes that the $\mathbb{Z}_3$ pattern creates a physically meaningful internal contrast — not merely a labelling convention — and motivates the coherence ratio used in Step 7. The observable BAO amplitude A_{Z_3} is derived independently in Step 7 from the ratio β_{coh}/β.

Step 1: Moment sums.

$$I_{\text{counter}} = \sum_{k=0,3,\ldots,30} \varphi^{-2k} = 1.059017$$
$$I_{\text{co}} = \sum_{k=1,2,4,\ldots,32} \varphi^{-2k} = 0.559017$$

Step 2: Enhancement factor (exact identity).

$$E = \frac{I_{\text{counter}} + I_{\text{co}}}{|I_{\text{counter}} - I_{\text{co}}|} = \frac{1.618034}{0.500000} = 3.236068 = \varphi^3 - 1 \quad \text{(exact)} \tag{88}$$

Step 3: Fractional energy modulation. The $\mathbb{Z}_3$ structure modulates energy density by $\Delta\rho/\rho = E/(1+E) = 76.39\%$.

Step 4: Z_3 fraction. Of the $N = 33$ shells, 11 are counter-rotating ($k \equiv 0 \bmod 3$), giving $f_{Z_3} = 11/33 = 1/3$.

Step 5: Geometric reduction factor. The $\mathbb{RP}^4$ antipodal symmetry halves the effective modulation: $f_{\text{sym}} = 1/2$.

Step 6: Intrinsic amplitude.

$$A_{\text{int}} = f_{\text{sym}} \times \frac{\Delta\rho}{\rho} \times f_{Z_3} = \frac{1}{2} \times 0.7639 \times \frac{1}{3} = 12.73\% \tag{89}$$

Note on the relationship between Steps 1–6 and Step 7. Steps 1–6 characterise the structural *energy-density* asymmetry of the $\mathbb{Z}_3$ shell system: the 12.73% figure is the fractional contrast in vacuum energy density between counter- and co-rotating shells, arising from the moment imbalance $I_{\text{counter}}/I_{\text{co}}$ modulated by $\mathbb{RP}^4$ antipodal symmetry.

This is a statement about the internal energy architecture, not directly about distance observables. The physically observable BAO distance modulation is a distinct quantity, defined as the fractional standing-wave perturbation to comoving distances at the $\mathbb{Z}_3$ period. Step 7 derives this independently from the ratio β_{coh}/β, which measures how much of the total suppression is coherently phased. The two quantities are not expected to be numerically related by a simple factor: the 12.73% energy-density contrast is diluted by the full suppression chain $N\varphi^4$ in a way that is not captured by the scalar ratio $E/(1+E)$ alone.

Step 7: Observable amplitude from coherence ratio. The physically observable fraction is determined by the hierarchy between the coherence factor and the total suppression:

$$A_{Z_3} = \frac{\beta_{\text{coh}}}{\beta} = \frac{6\cos(\pi/23)}{6\cos(\pi/23) \times 33 \times \varphi^4} = \frac{1}{N\varphi^4} = \frac{1}{33 \times 6.854} = 0.442\% \tag{90}$$

The large intrinsic amplitude (12.73%) is diluted by the $N\varphi^4 = 226$ factor that also generates the total suppression β: the same mechanism that addresses the cosmological constant problem suppresses the $\mathbb{Z}_3$ oscillation amplitude to 0.442%. This is the value used throughout the paper.

MCMC validation: posterior mean $A_{Z_3} = 0.0046 \pm 0.0029$ from all 33 BAO measurements (Section 5.4), consistent with the prediction at -0.06σ.

A.4 Number of Tori: $N = 33$

What topology constrains. The octonionic Hopf fibration $S^7 \to S^4$ and SO(8) triality require $N = 3M$ for integer M, so N must be a multiple of 3. The SU(2) representation on the S^3 fibre additionally requires M to be odd (all SU(2) irrep dimensions $2j+1$ are odd), restricting N to odd multiples of 3:

$$N \in \{3,\ 9,\ 15,\ 21,\ 27,\ 33,\ 39,\ 45,\ \ldots\} \tag{91}$$

This is a necessary condition. It does not select $N = 33$ from the list; that selection requires additional input.

Uniqueness theorem: $N = 33$ from the S^4 harmonic spectrum. The S^4 representation count at angular momentum l is:

$$d(S^4, l) = \frac{(2l+3)(l+1)(l+2)}{6} \tag{92}$$

At $l = 4$: $d(S^4, 4) = 11 \times 5 \times 6/6 = 55 = F_{10}$, the tenth Fibonacci number. The **Fibonacci self-termination condition** requires $d(S^4, l)$ to be a Fibonacci number. Direct evaluation identifies all hits across $l \geq 1$: at $l = 1$, $d = 5 = F_5$; at $l = 4$, $d = 55 = F_{10}$; for $l = 2, 3$, $d \in \{14, 30\}$, neither Fibonacci. No solution exists beyond $l = 4$: $d(S^4, l)$ grows as $\frac{1}{3}l^3$ while Fibonacci numbers grow as $\varphi^l/\sqrt{5}$; since exponential growth dominates polynomial growth, $F_n > d(S^4, l)$ for all sufficiently large l (verified: at $l = 5$, $F_{10} = 55 < 91 = d(S^4, 5)$, but $F_{11} = 89 < 91$ and $F_{12} = 144 > 91$, and the gap widens monotonically thereafter). The non-trivial Fibonacci hits are therefore exactly $l = 1$ and $l = 4$.

Step 1 (S^4, no identification): $l = 1$ fires first, giving $d = 5 = F_5$, $M = 1$, $N = 3$, and $r_s = R_H\varphi^{-4} = 649$ Mpc.

Step 2 ($\mathbb{RP}^4$ identification): The $\mathbb{Z}_2$ quotient projects out all odd-l harmonics ($Y_l(-x) = (-1)^l Y_l(x)$, so odd-l modes vanish). The $l = 1$ hit is forbidden. The first accessible Fibonacci hit is $l = 4$: $d(S^4, 4) = 55 = F_{10}$. From $l = 4$: $M = F_{10}/5 = 11$, $N = 3M = \mathbf{33}$. Simultaneously, the eigenvalue $l(l+3) = 28/R^2$ gives $k_{\text{BAO}} = l + 3 = 7$, so $r_s^{\text{IHC}} = R_H \varphi^{-7}$ is co-derived from the same harmonic.

Why the $\mathbb{RP}^4$ identification is necessary. The preceding derivation uses the S^4 harmonic dimension $d(S^4, l)$, but the topology of the manifold determines *which values of l are physically accessible*. On $\mathbb{RP}^4 = S^4/\mathbb{Z}_2$, the $\mathbb{Z}_2$ quotient projects out all odd-l harmonics (since $Y_l(-x) = (-1)^l Y_l(x)$). The physically accessible modes are $l \in \{0, 2, 4, 6, \ldots\}$ only.

On S^4 (no identification), the first Fibonacci hit is $l = 1$: $d(S^4, 1) = 5 = F_5$. This gives $N = 3$, $k_{\text{BAO}} = 4$, and $r_s = R_H \varphi^{-4} = 649$ Mpc — a factor of 4.4 above observation. On $\mathbb{RP}^4$, the $l = 1$ mode is forbidden; the first accessible Fibonacci hit is $l = 4$, giving the correct prediction. The topology is therefore not a free modelling choice: it is required by the data. Table 17 summarises the contrast.

Table 17: Observational consequence of $\mathbb{RP}^4$ vs S^4 topology. The $\mathbb{Z}_2$ identification is not aesthetic; without it, IHC predicts a BAO scale 4.4 times too large.

	S^4 (no identification)	$\mathbb{RP}^4 = S^4/\mathbb{Z}_2$
First Fibonacci hit	$l = 1$, $d = 5 = F_5$	$l = 4$, $d = 55 = F_{10}$
M, N	$M = 1$, $N = 3$	$M = 11$, $N = 33$
k_{BAO}	4	7
r_s prediction	649 Mpc	153.2 Mpc
vs. observed	$+\mathbf{339}\%$ error	$< 3.7\%$ (corrected by ξ)

The full derivation of $\mathbb{RP}^4$ from the inversion principle, including the Euler characteristic uniqueness proof, is given in Section 2.2.

Step 1: SO(8) triality. The octonionic Hopf fibration $S^7 \to S^4$ requires $N = 3M$, where M is the SU(2) representation dimension.

Step 2: $N = 3 \times 11 = 33$, derived from $d(S^4, 4) = 55 = F_{10}$ with $M = F_{10}/5 = 11$.

Step 3: SU(2) on the S^3 fibre. With $N = 33$ giving $M = 11$ counter-rotating groups, SO(8) transitivity forces irreducibility, and $j_{\max} = 5$ is the unique SU(2) irrep of dimension 11:

$$\underbrace{d(S^4, 4) = 55 = F_{10}}_{\text{uniqueness theorem}} \xrightarrow{\mathbb{Z}_3} \underbrace{M = 11}_{\text{counter-rot.}} \xrightarrow{\text{SO(8) transitivity}} \underbrace{\text{irrep, dim} = 11}_{\text{unique}} \xrightarrow{\text{SU(2)}} \underbrace{j_{\max} = 5}_{\text{unique}} \tag{93}$$

Step 4: Fibonacci validation. $F_{34}/F_{33} = 5{,}702{,}887/3{,}524{,}578 = 1.618033988749859$; relative error $= 2.22 \times 10^{-14}$. For volume scaling $V_k \propto \varphi^{-3k}$:

$$\frac{\sum_{i=0}^{33} \varphi^{-3i}}{\sum_{i=0}^{\infty} \varphi^{-3i}} > 99.9999\%$$

Consistency check: the series is well-converged at $N = 33$, confirming that additional shells contribute negligibly to physical observables.

A.5 Summary of Predictions

From $(\varphi, N = 33, R_H = c/H_0)$:

$$\Omega_\Lambda = 0.6889 \qquad \text{(from } \beta = 1345\text{)} \tag{94}$$

$$r_s^{\rm IHC} = 153.2 \text{ Mpc} \qquad \text{(from } k = 7 \text{ shell at } R_H\text{)} \tag{95}$$

$$A_{Z_3} = 0.442\% \qquad \text{(from } \beta_{\rm coh}/\beta\text{)} \tag{96}$$

$$\lambda_{Z_3} = 404.4 \text{ Mpc} \qquad \text{(from } R_H/11\text{)} \tag{97}$$

Ω_Λ and $r_s^{\rm IHC}$ are geometrically derived from $(\varphi, N = 33, \mathbb{RP}^4)$; neither is fitted to observational data.

B Heat-Kernel Proof of the UV–IR Seesaw Coefficient

This appendix provides the complete proof that $\rho_\Lambda^2 = \frac{1}{2}\rho_{UV}|\rho_{IR}|$, with the $\frac{1}{2}$ coefficient derived from the $\mathbb{RP}^4$ topology and confirmed by the $\mathbb{Z}_3$ shell self-consistency condition.

Step A: The $\mathbb{RP}^4$ heat kernel. $\mathbb{RP}^4 = S^4/\mathbb{Z}_2$ where $\mathbb{Z}_2 : x \to -x$. For a spinor field with anti-periodic boundary condition $\psi(-x) = -\psi(x)$, the method of images [10] gives the heat kernel on $\mathbb{RP}^4$ as a sum over the two elements $g \in \mathbb{Z}_2$, weighted by the spinor representation character $\chi_{\rm spinor}$:

$$K_{\mathbb{RP}^4}(x, x; t) = \frac{1}{|\mathbb{Z}_2|} \sum_{g \in \mathbb{Z}_2} \chi_{\rm spinor}(g)\, K_{S^4}(x, g \cdot x; t) = \frac{1}{2}\big[K_{S^4}(x, x; t) - K_{S^4}(x, -x; t)\big] \tag{98}$$

where $\chi_{\rm spinor}(\text{identity}) = +1$ and $\chi_{\rm spinor}(\text{antipodal}) = -1$ for anti-periodic spinors. The overall factor $1/|\mathbb{Z}_2| = 1/2$ is a topological constant, exact for any $\mathbb{RP}^n = S^n/\mathbb{Z}_2$.

Step B: Vacuum energy density and the observable cross-term. Integrating (98) over proper time t and dividing by the spatial volume gives:

$$\rho_{\rm vac} = \tfrac{1}{2}\big(\rho_{UV}^{\rm bare} - |\rho_{IR}|\big) \tag{99}$$

where $\rho_{UV}^{\rm bare}$ is UV-divergent (coincident term $K_{S^4}(x, x; t)$) and $|\rho_{IR}|$ is finite (antipodal term $K_{S^4}(x, -x; t)$, evaluated in Eq. 18). Squaring (99):

$$\rho_{\rm vac}^2 = \tfrac{1}{4}\big((\rho_{UV}^{\rm bare})^2 - 2\,\rho_{UV}^{\rm bare}\,|\rho_{IR}| + |\rho_{IR}|^2\big) \tag{100}$$

The three terms differ in their renormalisation properties: $(\rho_{UV}^{\rm bare})^2$ is scheme-dependent and absorbed into the cosmological constant counterterm $\delta\Lambda$ in the Einstein–Hilbert action $S \supset \int d^4x\sqrt{g}\,(M_{\rm Pl}^2 R/2 - \delta\Lambda)$; as a scheme-dependent divergent contribution it carries no physical content and its removal is the standard renormalisation of the bare cosmological constant. $|\rho_{IR}|^2 \ll \rho_\Lambda^2$ is negligible at physical scales. The cross-term $\frac{1}{4} \times 2\,\rho_{UV}^{\rm bare}\,|\rho_{IR}|$ is *scheme-independent*: $|\rho_{IR}|$ is the finite Casimir energy of Eq. (18), a physical observable set by R_H. A counterterm that could remove it would be proportional to R_H^{-4}, i.e. a dimension-8 operator suppressed by $M_{\rm Pl}^4$ — not present in the Einstein–Hilbert action. Any renormalisation that shifts $\rho_{UV}^{\rm bare}$ therefore leaves $|\rho_{IR}|$ unchanged, making their product $\rho_{UV}|\rho_{IR}| \equiv \mathcal{I}$ a renormalisation invariant of the $\mathbb{RP}^4$ vacuum.

$$\mathcal{I} \equiv \rho_{UV}\,|\rho_{IR}| \quad \text{is a renormalisation invariant,} \tag{101}$$

where ρ_{UV} denotes the scheme-independent renormalised UV scale.

Step C: The physical dark energy. After removing the scheme-dependent terms, the surviving scheme-independent content of (100) is the cross-term:

$$\rho_\Lambda^2 = \frac{1}{4} \times 2\,\rho_{UV}\,|\rho_{IR}| = \frac{1}{2}\,\rho_{UV}\,|\rho_{IR}| \tag{102}$$

The coefficient $\frac{1}{2}$ arises from two factors that together give the exact result: the orbifold factor $1/|\mathbb{Z}_2| = \frac{1}{2}$ from Step A enters squared (as the prefactor of ρ_{vac} in Eq. 99), giving $(1/2)^2 = 1/4$ in Eq. (100); the factor of 2 in the cross-term of the squared binomial then restores one power: $\frac{1}{4} \times 2 = \frac{1}{2}$. Both factors are exact and independent of any shell structure.

Step D: $|L_{\text{net}}| = 1/2$ as an algebraic identity. The $\mathbb{Z}_3$ imbalance $L_{\text{net}} = I_{\text{co}} - I_{\text{counter}}$ can be evaluated exactly for the φ-scaled geometric series in the limit $N \to \infty$ (corrections are $O(\varphi^{-2N})$, negligible for $N = 33$). Using $\varphi^6 = 8\varphi + 5$ (from $\varphi^2 = \varphi + 1$):

$$I_{\text{total}} = \frac{1}{1 - \varphi^{-2}} = \varphi \tag{103}$$

$$I_{\text{counter}} = \frac{1}{1 - \varphi^{-6}} = \frac{\varphi^6}{\varphi^6 - 1} = \frac{8\varphi + 5}{8\varphi + 4} \tag{104}$$

$$L_{\text{net}} = \varphi - \frac{2(8\varphi + 5)}{8\varphi + 4} = \frac{\varphi(8\varphi + 4) - (16\varphi + 10)}{8\varphi + 4} = \frac{-(2\varphi + 1)}{2(2\varphi + 1)} = -\frac{1}{2} \tag{105}$$

The factor $(2\varphi + 1)$ cancels exactly. This result holds for *any* φ-scaled $\mathbb{Z}_3$ shell structure with sufficiently large N; it is an algebraic identity of $\varphi^2 = \varphi + 1$, not a property specific to $N = 33$.

The significance is architectural: the $\mathbb{Z}_3$ shell structure built on φ-scaling *automatically* produces $|L_{\text{net}}| = 1/|\mathbb{Z}_2|$. The vacuum angular-momentum asymmetry matches the topological $\mathbb{Z}_2$ projection factor of $\mathbb{RP}^4$ as a consequence of the golden ratio identity $\varphi^2 = \varphi + 1$. The $\mathbb{Z}_2$, $\mathbb{Z}_3$, and φ structures are therefore mutually consistent at an algebraic level: the IHC architecture is self-consistent by construction.

Conclusion. Equations (101) and (102) together give the complete derivation of the seesaw:

$$\boxed{\rho_\Lambda^2 = \frac{1}{2}\,\rho_{UV}|\rho_{IR}|} \tag{106}$$

with no free parameters. The coefficient $\frac{1}{2}$ has a double origin: topologically, as the $1/|\mathbb{Z}_2|$ orbifold factor of $\mathbb{RP}^4 = S^4/\mathbb{Z}_2$; and algebraically, as the exact value of $|L_{\text{net}}|$ forced by $\varphi^2 = \varphi + 1$. Their equality is not a coincidence but a structural property of the IHC vacuum.

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AI-Assisted Tools

In accordance with MDPI editorial policy, the authors declare that large language model (LLM) tools were used for AI-assisted copy editing of this manuscript, including improvements to grammar, readability, and consistency of notation. All scientific content, derivations, numerical results, and conclusions are the sole work of the human authors. No LLM was used for autonomous content generation or to create novel scientific claims.

Author Contributions

Conceptualization, S.P.; Methodology, S.P. and L.H.; Formal analysis, S.P. and L.H.; Writing—original draft, S.P.; Writing—review and editing, S.P. and L.H. Both authors have read and agreed to the submitted version.

Data Availability

All BAO measurements used are publicly available from the BOSS DR12, eBOSS DR16, DESI DR2, WiggleZ, 6dFGS, and MGS data releases. Validation scripts and numerical results are archived at https://doi.org/10.5281/zenodo.19135785.

Conflicts of Interest

The authors declare no conflicts of interest.

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